Differential Equations Notes

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Fun Stuff

1. Course resource: https://ocw.mit.edu/courses/mathematics/18-03-differential-equations-spring
index.htm
2. Feynman Method: https://www.youtube.com/watch?v=FrNqSLPaZLc
3. Bad math writing: https://lionacademytutors.com/wp-content/uploads/2016/10/sat-math-section.
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4. Google AI experiments: https://experiments.withgoogle.com/ai
5. Babylonian tablet: https://www.maa.org/press/periodicals/convergence/the-best-known-old-babyl
6. Parabola in real world: https://en.wikipedia.org/wiki/Parabola#Parabolas_in_the_physical_
world
7. Parabolic death ray: https://www.youtube.com/watch?v=TtzRAjW6KO0
8. Parabolic solar power: https://www.youtube.com/watch?v=LMWIgwvbrcM
9. Robots: https://www.youtube.com/watch?v=mT3vfSQePcs, riding bike, kicked dog, cheetah, back-flip, box hockey stick
10. Cat or dog: https://www.datasciencecentral.com/profiles/blogs/dogs-vs-cats-image-classificat
11. History of logarithm: https://en.wikipedia.org/wiki/History_of_logarithms
12. Log transformation: https://en.wikipedia.org/wiki/Data_transformation_(statistics)
13. Log plot and population: https://www.google.com/publicdata/explore?ds=kf7tgg1uo9ude_&met_
y=population&hl=en&dl=en#!ctype=l&strail=false&bcs=d&nselm=h&met_y=population&scale_
y=lin&ind_y=false&rdim=country&idim=state:12000:06000:48000&ifdim=country&hl=en_US&dl=
en&ind=false

14. Yelp and NLP: https://github.com/skipgram/modern-nlp-in-python/blob/master/executable/Modern_NLP_in_Python.ipynb https://www.yelp.com/dataset/challenge

15. Polynomials and splines: https://www.youtube.com/watch?v=O0kyDKu8K-k, Yoda / matlab, https://www.google.com/search?q=pixar+animation+math+spline&espv=2&source=lnms&tbm=isch&sa=X&ved=0ahUKEwj474fQja7TAhUB3YMKHY8nBGYQ_AUIBigB&biw=1527&bih=873#tbm=isch&q=pixar+animatic mesh+spline, http://graphics.pixar.com/library/

16. Polynomials and pi/taylor series: Matlab/machin https://en.wikipedia.org/wiki/Chronology_of_computation_of_%CF%80 https://en.wikipedia.org/wiki/Approximations_of_%CF%80#Machin-lik formula https://en.wikipedia.org/wiki/William_Shanks

17. Deepfake: face https://www.youtube.com/watch?v=ohmajJTcpNk
dancing https://www.youtube.com/watch?v=PCBTZh41Ris

18. Pi digit calculations: https://en.wikipedia.org/wiki/Chronology_of_computation_of_%CF%80, poor shanks...https://en.wikipedia.org/wiki/William_Shanks

Course Introduction

1. What is a differential equation?

   (a) Example: Population growth
   $$\frac{dP}{dt} = 2P$$
   $P(t)$ is an unknown population function of time t. Many solutions, but can write the general solution. Given an initial population $P(0) = P_0$, we get a unique solution. Integration techniques give our solution in this case (analytic techniques).

   (b) General first order (derivative) equation: Notice we focus on change, not static values like an algebraic equation.
   $$\frac{dy}{dt} = f(t, y)$$
   $f(t, y)$ is some given function. Generally no technique can solve, so we will need other ideas (numerical and qualitative analysis).

   (c) ODE vs PDE

2. Why study differential equations?

   (a) Goal is to model the real world which involves making assumptions, building equations, and analysis to hopefully find solutions.

   (b) Useful in all sciences (population growth, disease spread, fluid dynamics, heat transfer, ecology, circuits, chemical reaction, stock market, etc)

   (c) List is long: https://en.wikipedia.org/wiki/List_of_named_differential_equations https://en.wikipedia.org/wiki/List_of_nonlinear_partial_differential_equations

   (d) Trouble is almost none can be solved, instead they are analyzed in general (theory) and approximated via simulations (coding).

3. Course topics:

   (a) Building (analyzing) mathematical models (general theme)

   (b) Ch1: First-order differential equations

   (c) Ch2: First order systems

   (d) Ch3: Linear systems

(e) Ch4: Forcing and resonance

(f) Ch5: Nonlinear systems

(g) Ch6: Laplace transform

Chapter 1: First-order differential equations

.1 1.1: Modeling via Differential Equations

1. Mathematical Modeling

   (a) A **mathematical model** translates a real-world problem into mathematics.

   (b) Assumptions simplify reality and determine the model.

   (c) More assumptions ⇒ simpler mathematics, but often a less realistic model.

   (d) The goal is to **understand, explain, and predict** behavior—not merely solve an equation.

   (e) Key steps:

      i. State assumptions which relate key quianties of study

      ii. Describe variables, parameters, and functions precisely

      iii. Combine (i) and (ii) into an equation to solve or study

      iv. Interpret (iii), state limitations, iterate as needed

2. Ordinary Differential Equations (ODEs)

   (a) Population Growth (walk thru above steps)
   Assumption: The growth rate is proportional to the population.

$$\frac{dP}{dt} = kP$$

   - First-order linear ODE.
   - Unknown: the function $P(t)$.
   - General solution:

$$P(t) = Ce^{kt}.$$

   - An initial condition, e.g. $P(0) = P_0$, gives a unique (particular) solution.
   - The parameter k represents the growth rate. This needs to be determined via empirical rule or data.
   - We will study:
     - analytical solutions (formula for P),
     - qualitative behavior (effects of different P_0, draw various cases, note equilibrium case),
     - numerical approximations (if analytical solutions cannot be found, we will use Python for this).
   - Discuss qualitative behavior for this one, equilibrium solutions, increasing and decreasing. Can reason out all possible solutions without formula.

   (b) Logistic Growth (refinement of above)
   Assumption: Growth slows as the population approaches a carrying capacity N.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) = f(P)$$

   - First-order nonlinear ODE.
   - Equilibrium solutions occur where $f(P) = 0$.

- Graph $f(P)$ to determine increasing/decreasing solutions.
- More realistic than exponential growth, though more complicated.

(c) General First-Order ODE

$$\frac{dy}{dt} = f(t, y)$$

- Describes how y changes.
- Much of this course studies equations of this form.
- Initial value problem case (IVP).
- Independent variable: usually time t.
- Dependent variable: the unknown function, e.g. $y(t)$.
- Variables vs. parameters
  - Variables change during the problem (t, y).
  - Parameters are fixed constants that describe the system (k, N, ...).
  - Changing a parameter gives a different model.

3. Systems of ODEs

(a) Predator–Prey Model

- Many systems involve several interacting variables.
- Let $R(t)$ = rabbits and $F(t)$ = foxes.

Assumptions:

- Rabbits reproduce naturally.
- Rabbits decrease through predation.
- Foxes die naturally.
- Foxes increase by consuming rabbits.

$$\begin{cases} \dfrac{dR}{dt} = \alpha R - \beta RF, \\ \dfrac{dF}{dt} = -\gamma F + \delta RF. \end{cases}$$

- First-order, coupled nonlinear system.
- This gives a coupled nonlinear system and equilibrium solutions have new interest. Play around with inc/dec rabbit/fox populations.
- Variables cannot be solved independently.
- We will study equilibria, stability, and long-term behavior.
- Linear systems can often be analyzed using linear algebra (linear approximations).

.2 1.2: Analytic technique: separation of variables

1. Separable equations

(a) There are few nice cases of first order DEs which we can completely solve by hand. These are important to gain insight from.

(b) Separable equation: f can be separated as a product of a function of t and function of y.

$$\frac{dy}{dt} = f(t, y) = g(t)h(y)$$

(c) Once separated the solution can be found via direct integration (if doable). Our focus is then just on calculus.

2. Examples:

(a) $\dfrac{dy}{dt} = 2ty$ is separable. How to find y?

$$\begin{aligned}
\frac{dy}{dt} &= 2ty \\
\int \frac{1}{y}\frac{dy}{dt}\, dt &= \int 2t\, dt \\
\int \frac{1}{y}\, dy &= \int 2t\, dt \\
\ln|y| &= t^2 + \hat{C} \\
|y| &= e^{t^2+\hat{C}} \\
y &= Ce^{t^2} \quad \text{(constant } C \text{ handles the sign)}
\end{aligned}$$

Note the use of the chain rule. We will usually skip this step. Now we see where the family of solutions comes from (integration constant). Given an initial condition $y(0) = y_0$ gives

$$y = y_0 e^{t^2}.$$

If we had an initial value problem of $y(0) = y_0$, we also could have handled as a definite integral from t_0 to t giving

$$\int_{y_0}^{y} \frac{1}{y}\, dy = \int_{t_0}^{t} 2t\, dt \quad \Rightarrow \quad y = y_0 e^{t^2}.$$

Note, solutions are easy to verify via the original equation.

(b) Exponential growth: An autonomous equation of the form $\frac{dy}{dt} = f(y)$. The right hand side depends on y alone.

$$\frac{dy}{dt} = ky$$

(c) Logistic growth: Another autonomous equation.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right)$$

Let's simplify things and take $k = 1$, $N = 1$.

$$\frac{dP}{dt} = P\,(1 - P) = f(P)$$

Again we are separable, but note the right hand side only depends on P. This is called an *autonomous* equation. We solve the same way. Note the equilibrium solutions $P = 0, 1$ as we

divide.

$$\begin{aligned}
\frac{dP}{dt} &= P\,(1-P) \\
\int \frac{1}{P(1-P)}\,dP &= \int\, dt \\
\int \frac{1}{P} + \frac{1}{1-P}\,dP &= \int\, dt \\
\ln|P| - \ln|1-P| &= t + \hat{C} \\
\ln\left|\frac{P}{1-P}\right| &= t + \hat{C} \\
\frac{P}{1-P} &= e^{t+\hat{C}} \\
\frac{P}{1-P} &= Ce^{t} \\
P &= (1-P)Ce^{t} \\
P &= \frac{Ce^{t}}{Ce^{t}+1} = \frac{1}{1+Ae^{-t}}
\end{aligned}$$

Note the use of partial fraction decomposition. For $A = 1$ this is the sigmoid function (logistic curve). For the original equation with rate k and carrying capacity N, we get

$$P = \frac{NP_0e^{kt}}{N + P_0(e^{kt}-1)}.$$

3. Difficulties with separation:

   (a) Implicit solutions, and missing solutions.

   $$\frac{dy}{dt} = \frac{y}{1+y^2}$$

   has solution

   $$\ln|y| + \frac{y^2}{2} = t + C$$

   and it is not possible to solve for y (known as an implicit solution). Like the equation of a circle, this still completely describes the solution set. We cannot write y explicitly as a function of t. Also, $y = 0$ is an *equilibrium solution* which we missed due to (zero) division, yet our implicit solution does not make sense for $y = 0$. Further, there are examples where integration is not possible. Don't forget to check for equilibrium solutions!

   (b) Integration is not always possible.

4. Mixing problems: A 200 gal tub contains pure water. Sugar water is pumped in at 5 tablespoons sugar per gallon at a rate of 3 gals per minute. The take is mixed and outflow rate is the same. How much sugar is in the tank after 10 mins? What is the equilibrium solution? Setting up the equation is the challenge. Build it as

   $$\text{Rate of change} = \text{Rate in} - \text{Rate out}$$

   writing concentrations as $tbspsalt/gal$ and rates as gal/min. The product of these gives our salt rate of change. Then for $S(t)$ in tbsp of salt and t in minutes,

   $$\frac{dS}{dt} = 5 \cdot 3 - \frac{S}{200} \cdot 3$$

   This is a separable equation yet again. Ex 39 also works.

.3 1.3: Qualitative technique: slope fields

1. dfeild software: `https://math.rice.edu/~dfield/dfpp.html`
2. Analytic solutions are often impossible (or difficult) to obtain. Instead, we can study a differential equation qualitatively.
   - Visualize solutions using a *slope field* (or *direction field*).
   - Approximate solutions numerically.
   - Predict long-term behavior (equilibria and stability).
3. Slope fields visualize the differential equation
   $$\frac{dy}{dt} = f(t, y).$$
   At each point (t, y), draw a short line segment with slope
   $$m = f(t, y).$$
   Solution curves are tangent to these line segments everywhere.
4. Example: $y' = 2y, \qquad y(0) = 1$
   - Slopes depend only on y (autonomous equation), so each horizontal row has identical slope.
   - Slopes are positive above $y = 0$, negative below, and zero on $y = 0$.
   - Sketch the solution through $(0, 1)$.
   - Compare with the exact solution
     $$y = e^{2t}.$$
5. Example:
   $$y' = 2 - y.$$
   - Nullcline: $y = 2$.
   - Above $y = 2$, slopes are negative.
   - Below $y = 2$, slopes are positive.
   - Every nearby solution moves toward $y = 2$, so it is a stable equilibrium.
   - Exact solution:
     $$y = 2 + Ce^{-t}.$$
6. Isoclines
   - An isocline is a curve where the slope is constant:
     $$f(t, y) = s.$$
   - The nullcline is the special isocline where
     $$f(t, y) = 0.$$
   - Plotting several isoclines makes hand sketches much easier.
7. Example:
   $$P' = P(1 - P).$$
   - Nullclines: $P = 0$ and $P = 1$.
   - Slopes are positive for $0 < P < 1$.

- Slopes are negative elsewhere.
- Therefore
  - $P = 1$ is stable.
  - $P = 0$ is unstable.
- Compare with the familiar logistic curve.

8. Example:
$$y' = 1 + t - y.$$
   - Nullcline:
$$y = t + 1.$$
   - Isoclines:
$$y = t + 1 + s.$$
   - Unlike autonomous equations, the nullcline is not horizontal.
   - Solutions approach the line
$$y = t,$$
     which is an exact (nonconstant) solution.

9. Special case:
$$y' = f(t).$$
   - Slopes depend only on t.
   - Every vertical column has the same slope.
   - Isoclines are vertical lines.
   - Example:
$$y' = 2t,$$
     whose solutions are
$$y = t^2 + C.$$

10. Using `dfield`
    - Computes slope fields by evaluating $f(t, y)$ on a grid.
    - Numerically approximates solution curves from chosen initial conditions.
    - Useful for exploring equations before attempting analytic methods.

11. Key takeaways
    - A slope field visualizes the differential equation itself.
    - Solution curves are everywhere tangent to the direction arrows.
    - Isoclines organize regions of equal slope; nullclines identify zero slope.
    - Autonomous equations have horizontal isoclines and equilibrium solutions.
    - Qualitative analysis reveals long-term behavior even when no explicit solution exists.

12. A differential equation tells us how a solution changes, not what the solution is. A slope field converts this local information into a global picture of all possible solutions.

.4 1.4: Numerical technique: Euler's method

1. For most differential equations, analytic solutions are not possible. We need numerical methods.
   - We trade exact for approximate, though as close to the exact solution as desired if we are persistent.
   - The simplest idea harnesses tangent lines (just like direction fields) and is called Euler's method.
   - Drawback with this simple idea will be inefficiency, but this can be improved with more advanced methods which can be derived and analyzed via calculus ideas (Taylor series, numerical analysis).

2. Euler's method for approximating solutions to $\dfrac{dy}{dt} = f(t, y)$:

   (a) Idea: Approximate the derivative (tangent line slope) with a difference quotient (secant line slope):
   $$\frac{df}{dx} = \lim_{h\to 0} \frac{\Delta f}{\Delta x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{\Delta x\to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$
   translated to
   $$\frac{dy}{dt} = \lim_{\Delta t\to 0} \frac{y(t+\Delta t) - f(t)}{\Delta t} \approx \frac{y(t+\Delta t) - y(t)}{\Delta t}$$
   for some t and Δt. Fitting into our equation,
   $$\frac{y(t+\Delta t) - y(t)}{\Delta t} \approx f(t, y)$$
   gives a way to approximate the future
   $$y(t+\Delta t) \approx y(t) + \Delta t f(t, y).$$

   (b) First step: $y_1 = y_0 + \Delta t f_0$
   - Notation: $y(0) = y_0$ (given initial value), $y_1 \approx y(t_1)$, $t_1 = t_0 + \Delta t$ (some chosen time step), $f_0 = f(t_0, y_0)$ (slope of the solution curve at t_0). Likewise for t_n, y_n and f_n.
   - Follow the tangent line to get the first approximation:
   $$y_1 = y_0 + \Delta t f_0$$
   - Example: $y' = 2y$ with $t_0 = 0, y(t_0) = 1$. Then the slope of the tangent is $f_0 = 2$ and the tangent line is $y = y_0 + f_0(t - t_0)$ giving
   $$y_1 = y_0 + \Delta t f_0 = 1 + 2\Delta t.$$
   Graph with the true solution $y(t) = e^{2t}$. Note we cannot take Δt too big else we will stray far from the curve.

   (c) General nth time step: $y_{n+1} = y_n + \Delta t f_n$
   - Following the second tangent line in above example gives $y_2 = y_1 + \Delta t f_1$. In general,
   $$y_{n+1} = y_n + \Delta t f_n \quad \left(\text{rearranged as } \frac{y_{n+1} - y_n}{\Delta t} = f_n\right)$$
   - Example: Returning to $y' = 2y$, we get
   $$\begin{aligned} y_{n+1} &= y_n + \Delta t f_n \\ &= y_n + \Delta t 2 y_n \\ &= (1 + 2\Delta t) y_n \\ &= (1 + 2\Delta t)^2 y_{n-1} \\ &= \vdots \\ &= (1 + 2\Delta t)^n y_0 \end{aligned}$$

If $y(0) = 1$, say we are truly after an approximation to $y(1)$. Then if we take $n+1$ time steps ($\Delta t = \frac{1}{n}$)

$$y_{n+1} = \left(1 + \frac{2}{n}\right)^n \approx e^2 \quad \left(\text{recall } (1 + \frac{1}{n})^n \to e\right)$$

and it looks like it is working!

- Taking smaller steps (Δt) results in more steps needed (larger n) to get to end time T. So we converge to the true solution.
- Draw on $y' = 2y$ example 2 steps to $T = 1$ via 4 steps. Close approximation in the end, but how much better?

3. General Euler's method algorithm: Given the initial value problem

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0,$$

and a chosen step size Δt:

(a) Set

$$t = t_0, \qquad y = y_0.$$

(b) Repeat until the desired final time is reached:

$$\boxed{\begin{aligned} &t_{n+1} = t_n + \Delta t, \\ &y_{n+1} = y_n + \Delta t\, f(t_n, y_n), \end{aligned} \qquad n = 0, 1, 2, \ldots}$$

(c) The computed values

$$(t_0, y_0),\ (t_1, y_1),\ \ldots,\ (t_n, y_n)$$

approximate the true solution at the times

$$t_n = t_0 + n\Delta t.$$

4. Example: Random homework problem.

5. Idea of numerical analysis:

(a) Build mathematical guarantees about a numerical method such as Euler's method (simplest explicit method).

(b) Local error (per iteration), global error (accumulation of local errors)

(c) Convergence, rate of convergence (under suitable conditions, as $\delta t \to 0$ we converge to the true solution, rate of Euler's method is first order accurate, halving Δt roughly halves the error)

(d) Stability analysis (conditions for convergence)

(e) Higher order methods (more terms, better approximation, faster convergence)

(f) Key tool: Taylor series and Taylor's Theorem (Chapter 7 holds mathematical analysis)

6. Programming assignment: Part 1, reproduce results in PDF. Details on Canvas.

.5 1.5: Existence and uniqueness of solutions

1. **The basic question:** Given an initial value problem, can we be sure that a solution exists? If it exists, is it the *only* solution?

   These are theoretical questions, but they have practical consequences. If a differential equation is being used as a model, we want to know that the model actually determines a solution. This is also important when using numerical methods: if the IVP does not have a well-defined solution, there is nothing for a numerical method to approximate.

2. **We have seen questions like this before.**

   (a) Quadratic equations: How many solutions does

   $$ax^2 + bx + c = 0$$

   have? How do we know?

   (b) Linear systems: When does

   $$A\vec{x} = \vec{b}$$

   have a solution? When is the solution unique?

   $$\det(A) \neq 0, \qquad \text{full rank}, \qquad \text{linearly independent columns}, \ldots$$

   (c) Differential equations: Given

   $$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0,$$

   when can we be sure that a solution exists? When is it unique?

3. **What can go wrong?**

   Consider an initial value problem. We might encounter:

   - no solution;
   - more than one solution;
   - a solution that exists only for a limited range of t.

   The goal of this section is to identify conditions that rule out these problems.

4. **Why does uniqueness matter?**

   Consider the differential equation

   $$\frac{dy}{dt} = y(1 - y).$$

   There are two equilibrium solutions:

   $$y(t) = 0, \qquad y(t) = 1.$$

   Suppose a solution starts at $y(0) = \frac{1}{2}$. Since

   $$y(1 - y) > 0 \qquad \text{for } 0 < y < 1,$$

   the solution initially increases.

   Could this solution ever cross $y = 1$?

   If it did, at the moment it reached $y = 1$ there would be two different solutions through the same point: our solution and the equilibrium solution $y = 1$.

   Thus, if solutions are unique, **solution curves cannot cross**. In particular,

   $$0 < y(t) < 1$$

   for all t for which the solution exists.

   This illustrates the practical importance of uniqueness: it lets us draw conclusions about the behavior of solutions without solving the differential equation.

5. **Existence:** Consider the IVP
$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0.$$
The basic idea is that f needs to be "nice enough" near the initial point.

   (a) **Existence Theorem:** Suppose $f(t, y)$ is continuous on a rectangle
   $$R = \{(t, y) : a < t < b,\ c < y < d\}.$$
   If (t_0, y_0) is in R, then there exists a solution $y(t)$ to the IVP for t sufficiently close to t_0.

   (b) "Sufficiently close" means that there is some $\epsilon > 0$ such that the solution exists for
   $$|t - t_0| < \epsilon.$$
   The theorem guarantees existence *locally*; it does not necessarily guarantee that the solution exists for all time.

   (c) If f is not continuous near (t_0, y_0), this theorem does not apply. A solution may or may not exist.

   (d) Example:
   $$\frac{dy}{dt} = 1 + y^2, \qquad y(0) = 0.$$
   The solution is
   $$y(t) = \tan(t).$$
   Thus a solution exists near $t = 0$, as the theorem predicts, but it blows up at
   $$t = \frac{\pi}{2}.$$
   Increasing y generates a quadratic increase in $\frac{dy}{dt}$ which gets out of hand. So existence near the initial point does not imply existence for all t.

6. **Uniqueness:** Existence is only part of the story. Suppose we know that a solution exists. Could there be several different solutions passing through the same initial point?

   (a) **Existence and Uniqueness Theorem:** Suppose $f(t, y)$ and
   $$\frac{\partial f}{\partial y}(t, y)$$
   are continuous on a rectangle R containing (t_0, y_0). Then the IVP has a unique solution for t sufficiently close to t_0.

   (b) The condition on $\partial f/\partial y$ gives us a way to guarantee that two different solution curves cannot pass through the same initial point.

   (c) **Existence does not guarantee uniqueness.** Consider
   $$\frac{dy}{dt} = 3y^{2/3}, \qquad y(0) = 0.$$
   Here $f(y) = 3y^{2/3}$ is continuous, so the existence theorem guarantees a solution. However, the solution is not unique. For example,
   $$y(t) = 0 \qquad \text{and} \qquad y(t) = t^3$$
   both satisfy the IVP. In fact, a solution can remain at 0 for any amount of time and then depart:
   $$y(t) = \begin{cases} 0, & t \le C, \\ (t - C)^3, & t > C. \end{cases}$$

The problem is that

$$\frac{\partial f}{\partial y} = 2y^{-1/3}$$

is not continuous at $y = 0$, so the uniqueness theorem does not apply.

(d) This example shows why existence and uniqueness are separate questions:

- f is continuous $\Rightarrow$ existence is guaranteed;
- f_y is not continuous at the initial point $\Rightarrow$ uniqueness is not guaranteed.

Thus, "a solution exists" does not necessarily mean "the initial condition determines a single solution."

(e) Example:

$$\frac{dy}{dt} = \frac{t}{y}, \qquad y(0) = 0.$$

Here $f(t, y) = t/y$ is not defined at the initial point. For $t \neq 0$, however, both

$$y(t) = t \qquad \text{and} \qquad y(t) = -t$$

satisfy the differential equation, and both approach $y = 0$ as $t \to 0$.

Thus the point $(0, 0)$ lies on multiple solution curves of the differential equation, but strictly speaking these are not classical solutions of the IVP on an interval containing $t = 0$, since the differential equation is undefined at the initial point.

This illustrates why the hypotheses of the theorem matter: if f is not defined at the initial point, we cannot use the theorem to guarantee existence or uniqueness.

7. **Why uniqueness matters:** Suppose f is nice enough that the existence and uniqueness theorem applies. Then:

   If two solution curves ever pass through the same point at the same time, they must be the same solution.

   This simple idea gives us a powerful way to understand solutions without solving the differential equation explicitly.

8. **Example: the logistic equation**

   Consider

$$\frac{dy}{dt} = y(1 - y).$$

   The constant functions

$$y(t) = 0, \qquad y(t) = 1$$

   are equilibrium solutions.

   Suppose

$$y(0) = \frac{1}{2}.$$

   Since

$$0 < \frac{1}{2} < 1,$$

   and

$$f(y) = y(1 - y) > 0 \qquad \text{for } 0 < y < 1,$$

   the solution initially increases.

   Could it cross $y = 1$?

No. If it reached $y = 1$ at some time t_1, then there would be two solutions passing through the same point $(t_1, 1)$:

$$y(t) = 1$$

and our solution. Uniqueness says this is impossible unless the two solutions are identical. But our solution started at $y(0) = 1/2$, so they cannot be identical.

Similarly, the solution cannot cross $y = 0$.

Therefore,

$$0 < y(t) < 1$$

for all t for which the solution exists. Since $y'(t) > 0$ in this interval, the solution is always increasing.

This gives us qualitative information about the solution without solving the differential equation.

In fact, the solution approaches 1 as $t \to \infty$. If it approached some value $L < 1$, then

$$f(L) = L(1 - L) > 0,$$

so the solution would still have a positive slope near L and could not settle down there.

9. Depth: Avoiding future bad behavior (finite time blow up).
   - Fact: Continuity of f and $\frac{\partial f}{\partial y}$ at all points does not ensure that the solution will reach $t = \infty$.
   - Example: $\dfrac{dy}{dt} = y^2, y(0) = 1$ has continuous f as well as $\frac{\partial f}{\partial y} = 2y$. Separation of variables gives the solution
   $$y(t) = \frac{1}{1 - t}$$
   which blows up to ∞ at $t = 1$. To ensure controlled growth of the solution, we require more of f.
   - Theorem (big one for this chapter, maybe course?): If
   $$\left|\frac{\partial f}{\partial y}\right| \leq L$$
   for all t, y, then there is a unique solution for all $y(0)$ reaching all t. (Essentially we need to control the growth of f in y)
   - Let's prove this theorem. How do we show a solution exists when we have no formula for y or f? The answer is to construct a solution.
   - Example: Show off idea with $y' = y$, $y(0) = 1$ (know the solution is e^t). Construct solution y via a sequence of functions y_k:
   $$\begin{aligned} y_0' &= 0, y(0) = 1 \\ y_1' &= y_0 \Rightarrow y_1 = 1 + t \\ y_2' &= y_1 \Rightarrow y_2 = 1 + t + \frac{t^2}{2} \\ y_3' &= y_2 \Rightarrow y_3 = 1 + t + \frac{t^2}{2} + \frac{t^3}{6!} \\ &\vdots \end{aligned}$$
   generating the power series for solution e^t. Do need to show that series converges (ratio test). The proof of our theorem follows the sameish approach.
   - Proof: We construct our sequence as
   $$\text{(equation:)}\ \frac{dy_{n+1}}{dt} = f(t, y_n(t)), \quad \text{(solution:)}\ y_{n+1} = y_0 + \int_0^t f(s, y_n(s))\, ds$$

Then,

$$y_{n+1}(t) - y_n(t) = \int_0^t [f(s, y_n(s)) - f(s, y_{n-1}(s)) \, ds$$

When $|\partial f/\partial y| \leq L$, the difference $|f(y_n) - f(y_{n-1})| \leq L|y_n - y_{n-1}|$. Then,

$$|y_2 - y_1| \leq \int_0^t L|y_1 - y_0| \, ds \leq Lt|y_1 - y_0|_{max}$$

and also

$$|y_3 - y_2| \leq \int_0^t L|y_2 - y_1| \, ds \leq \int_0^t L^2 t|y_1 - y_0|_{max} \, ds = \frac{L^2 t^2}{2}|y_1 - y_0|_{max}$$

leading to

$$|y_n - y_{n-1}| \leq \frac{L^n t^n}{n!}|y_1 - y_0|_{max}$$

which rapidly approaches zero. Then for n larger and N larger,

$$|y_N - y_n| \leq |y_N - y_{N-1}| + |y_{N-1} - n_{N-2}| + \cdots + |y_{n+1} - y_n| \leq C\frac{L^n t^n}{n!}$$

which also approaches zero. This is a Cauchy sequence which implies y_n converges to some limit $y(t)$. This implies y_{n+1} does likewise giving

$$y_{n+1} = y_0 + \int_0^t f(s, y_n(s)) \, dt \quad \rightarrow \quad y(t) = y_1 + \int_0^t f(s, y(s) \, ds$$

solving $\frac{dy}{dt} = f(t, y)$.

.6 1.6: Equilibria and the phase line

1. Return to autonomous equations $\dfrac{dy}{dt} = f(y)$.

   (a) Example: Logistic equation $\dfrac{dy}{dt} = y(1 - y)$.

   - Already noted that direction fields of autonomous equations are constant on horizontal lines. Why? $f(t, y)$ does not depend on t.
   - A phase line compresses this information into a single line. It answers a basic question: given the current value of y, will the solution increase or decrease? Draw a vertical line and mark the equilibrium values. Arrows indicate the direction of motion. Already saw this in the previous sections, though this simplification helps us gain intuition more easily.
   - **Reading a phase line:** The arrows determine the long-term behavior of solutions without solving the differential equation. In particular, they tell us which equilibria are attracting and which are repelling.
   - An equilibrium y_0 is **stable** if solutions that start sufficiently close to y_0 remain close to it. It is **asymptotically stable** if they also approach y_0 as $t \to \infty$.

   (b) Steps to draw a phase line

   - Find equilibrium solutions, label on number line. Because f is continuous, these are the only places f can change sign assuming continuous f.
   - Find the sign of f between equilibrium solutions. This gives increase / decrease of solution y.

2. Classification of equilibrium points:

   (a) **Sink (stable):** Solutions approach y_0 from both sides as $t \to \infty$. If $f'(y_0) < 0$, then y_0 is a sink.

(b) **Source (unstable):** Solutions move away from y_0 on both sides as $t \to \infty$. If $f'(y_0) > 0$, then y_0 is a source.

(c) **Semistable (stable from one side):** Solutions approach y_0 from one side and move away from it on the other. If $f'(y_0) = 0$, the derivative test is inconclusive; inspect the signs of $f(y)$ on either side of y_0.

3. **Example:** Consider

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right)\left(\frac{P}{M} - 1\right), \qquad N > M > 0.$$

This is a modified logistic model with a *minimum sustainable population* (or threshold) M.

Choose $N = 10$, $M = 2$, and assume $k > 0$:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{10}\right)\left(\frac{P}{2} - 1\right).$$

(a) Find and classify the equilibrium solutions.

(b) Draw the phase line.

(c) What happens to a population starting at $P(0) = 1$? At $P(0) = 5$?

(d) Interpret the results biologically. What role does the threshold M play?

The equilibria are

$$P = 0, \qquad P = 2, \qquad P = 10.$$

For $k > 0$, the phase line is

$$0 \quad \longleftarrow \quad 2 \quad \longrightarrow \quad 10 \quad \longleftarrow .$$

Thus $P = 0$ and $P = 10$ are sinks, while $P = 2$ is a source. A population starting below the threshold $M = 2$ declines toward extinction, while a population starting above M grows toward the carrying capacity $N = 10$.

4. **Example:** Consider

$$\frac{dy}{dt} = y(y-1)^2.$$

(a) Find the equilibrium solutions.

(b) Draw the phase line.

(c) Classify each equilibrium as a sink, source, or semistable.

(d) Describe the long-term behavior for different initial conditions.

Notice that

$$f'(0) = f'(1) = 0,$$

so the derivative test is inconclusive. We must instead examine the sign of $f(y)$ on either side of each equilibrium.

5. **Connection to uniqueness:** If the hypotheses of the uniqueness theorem hold, solution trajectories cannot cross equilibrium solutions. Therefore, equilibrium solutions divide the phase line into regions in which solutions remain for all future time (as long as the solution exists).

.7 1.7: Bifurcations

1. **What happens when a parameter changes?**

   Here we study families of differential equations that depend on one or more parameters. A parameter can change the equilibria and, more importantly, the long-term behavior of solutions.

   - Example:
   $$\frac{dy}{dt} = ky,$$
   where k is a parameter. For $k < 0$, the equilibrium $y = 0$ is a sink; for $k > 0$, it is a source. At $k = 0$, every constant function is an equilibrium.
   - A *bifurcation* occurs when a change in a parameter causes a qualitative change in the behavior of solutions, such as a change in the number or stability of equilibria.

2. **Logistic growth with harvesting.**

   Consider
   $$\frac{dP}{dt} = 4P - P^2 - h = f(P),$$
   where $h > 0$ is a constant harvesting rate.

   (a) Equilibria occur where
   $$f(P) = 4P - P^2 - h = 0.$$
   Solving the quadratic gives
   $$P = 2 \pm \sqrt{4 - h}.$$

   (b) Explore the phase line for several values of h:
   - $h = 0$: two equilibria (no harvesting);
   - $h = 3$: two equilibria (light harvesting);
   - $h = 4$: one equilibrium (critical harvesting);
   - $h = 5$: no equilibria (overharvesting).

   (c) What happens as h increases? The two equilibria approach each other, merge at $h = 4$, and then disappear. Thus $h = 4$ is the *bifurcation value*.

   (d) We can find the bifurcation value without solving for the equilibria. At the critical value, the graph of $f(P)$ is tangent to the P-axis, so
   $$f(P) = 0, \qquad f'(P) = 0.$$
   For
   $$f(P) = 4P - P^2 - h,$$
   we have
   $$f'(P) = 4 - 2P.$$
   Thus $f'(P) = 0$ gives $P = 2$, and substituting into $f(P) = 0$ gives
   $$h = 4.$$

   (e) **Interpretation:** $h = 4$ is the maximum sustainable harvesting rate. For $h < 4$, there is a positive equilibrium population. At $h = 4$, the positive equilibria merge. For $h > 4$, no positive equilibrium exists, so the population eventually declines.

   (f) Draw the *bifurcation diagram*: plot equilibrium values P^* on the vertical axis against the parameter h on the horizontal axis. Indicate which equilibrium branches are stable and unstable.

3. **Another example:** Draw and interpret the bifurcation diagram for

$$\frac{dy}{dt} = Ay - y^2 = y(A - y).$$

   (a) Find the equilibrium solutions as functions of A.

   (b) Determine their stability for different values of A.

   (c) Identify the bifurcation value.

   (d) Draw and interpret the bifurcation diagram.

.8 1.8: Linear equations

1. We end the chapter with another important analytic technique. There are many techniques for solving differential equations, but two of the most widely useful for first-order equations are:

   (a) **Separation of variables**: separate the variables and integrate.

   (b) **Linear equations**: use the structure of the equation to construct the solution.

2. For linear equations, we will see two approaches:

   - **Undetermined coefficients**: make an educated guess for a particular solution.
   - **Integrating factors**: systematically compute a particular solution (next section).

3. **First order linear equations** can be written in the form:

$$\frac{dy}{dt} = a(t)y + b(t)$$

   where $a(t)$ and $b(t)$ are known functions of t. The important feature is that the unknown function $y(t)$ appears *linearly*: it only appears to the first power and is not multiplied by another function involving y. For example,

$$y' = t^2 y + \sin(t)$$

   is linear (in y on the RHS), while

$$y' = y^2 + t, \qquad y' = \frac{1}{y} + t, \qquad y' = \sin(y) + t$$

   are nonlinear.

   - If $b(t) = 0$, the equation is called **homogeneous** (or non-forced):

$$y' = a(t)y.$$

   - If $b(t) \neq 0$, the equation is called **nonhomogeneous** (or forced):

$$y' = a(t)y + b(t).$$

     The terminology "forced" comes from applications: $b(t)$ can represent an external input or forcing acting on the system.

4. Meaning of linearity:

   (a) Theorem: Linearity principle (homogeneous general solution): If $y_h(t)$ solves the homogeneous equation $\frac{dy}{dt} = a(t)y$, then any constant times y_h also solves it. Not surprising because this is a separable equation and we know:

$$\frac{dy}{dt} = a(t)y \quad \Rightarrow \quad \int \frac{1}{y} dy = \int a(t) dt \quad \Rightarrow \quad y_h(t) = Ce^{\int a(t)dt}$$

(b) Particular solution (to the nonhomogeneous equation): $y_p(t)$ solves the nonhomogeneous equation $\frac{dy}{dt} = a(t)y + b(t)$.

(c) Theorem: Extended linearity principle (nonhomogeneous general solution): For linear equations, adding these two parts gives the complete solution

$$y = y_h + y_p$$

Easy to plug in and see why.

(d) Note: This only works because of the linear term (y^1). If we instead had y^2 adding y_h^2 to y_p^2 would not give $(y_h + y_p)^2$.

5. Method for solving linear equations:

(a) Solve the homogeneous equation (separable equation) to get y_h.

(b) Find a particular solution y_p. Two techniques:

- Undetermined coefficients (guess and check)
- Integrating factor (next section)

(c) Add two solutions per previous theorem.

6. Method of undetermined coefficients:

(a) Example: $\dfrac{dy}{dt} = -2y + e^t$.

- The homogeneous equation is $y' = -2y$ with solution $y_h = Ce^{-2t}$.
- For the particular solution, we need
$$y' + 2y = e^t$$
Then y better be some multiple of e^t. Let
$$y_p = \alpha e^t$$
and find the α which works.
$$y_p' + 2y_p = \alpha e^t + 2\alpha e^t = 3\alpha e^t = e^t$$
Then $\alpha = 1/3$ and $y_p = \frac{1}{3}e^t$.
- The complete solution is then
$$y = y_h + y_p = Ce^{-2t} + \frac{1}{3}e^t$$

(b) This guessing really only works for nice functions for $b(t)$.

- e^t needs αe^t
- $\sin(t)$ (or $\cos(t)$) needs $\alpha \sin(t) + \beta \cos(t)$ (why both?) Example from HW.

(c) Of course guessing isn't a good technique: Example

$$y' = -2y + e^{-2t}$$

- Why doesn't $y_p = \alpha e^{-2t}$ work? It solves the homogeneous equation.
- What might work instead? $y_p = \alpha t e^{-2t}$ via the product rule.
- The next section trades guessing for computation (integration).

.9 1.9: Integrating factors for linear equations

1. This section uses a trick to integrate first order linear nonhomogeneous equations directly.
   - Advantage: Straightforward technique for finding analytic solutions.
   - Disadvantage: Integration is hard, even impossible much of the time.
2. Example: $\dfrac{dy}{dt} = -2y + e^t$ revisited.
   (a) Rearrange and note the similarity to the product rule.

$$\frac{dy}{dt} + 2y = e^t$$

   We need to multiply both sides by e^{2t} (called the *integrating factor*) in order to see the product rule exactly.

$$\begin{aligned} e^{2t}\frac{dy}{dt} + 2e^{2t}y &= e^{2t}e^t \\ \frac{d}{dt}\left(e^{2t}y\right) &= e^{3t} \\ e^{2t}y &= \int e^{3t}\ dt \\ e^{2t}y &= \frac{1}{3}e^{3t} + C \\ y &= \frac{1}{3}e^t + Ce^{-2t} \end{aligned}$$

   as we found in the last section. Trick is to force the product rule to appear.
3. Method of integrating factor: General first order equation

$$\frac{dy}{dt} = a(t)y + b(t) \quad \Rightarrow \quad \frac{dy}{dt} - a(t)y = b(t)$$

   (a) Rearrange, introduce exponential (integrating factor), then compact as product rule:

$$\begin{aligned} \frac{dy}{dt} - a(t)y &= b(t) \\ e^{-\int a(t)\ dt}\frac{dy}{dt} - a(t)e^{-\int a(t)\ dt}y &= b(t)e^{-\int a(t)\ dt} \\ \frac{d}{dt}\left(e^{-\int a(t)\ dt}y\right) &= b(t)e^{-\int a(t)\ dt} \\ \frac{d}{dt}\left(\mu(t)y\right) &= b(t)\mu(t) \\ \mu(t)y &= \int b(t)\mu(t)\ dt \\ y &= \frac{1}{\mu(t)}\int b(t)\mu(t)\ dt \end{aligned}$$

   Note, check the differentiation of the integrating factor $\mu(t) = e^{-\int a(t)\ dt}$:

$$\frac{d}{dt}e^{-\int a(t)\ dt} = e^{-\int a(t)\ dt}(-a(t))$$

   Better to execute this as a strategy rather than try to memorize the formula.

(b) Example: Solve $\frac{dy}{dt} = -\frac{y}{t} + 2, y(1) = 3$

$$\begin{aligned}
\frac{dy}{dt} &= -\frac{y}{t} + 2 \\
\frac{dy}{dt} + \frac{y}{t} &= 2 \\
e^{\int 1/t \ dt}\frac{dy}{dt} + e^{\int 1/t \ dt}\frac{1}{t}y &= 2e^{\int 1/t \ dt} \\
e^{\ln|t|}\frac{dy}{dt} + e^{\ln|t|}\frac{1}{t}y &= 2e^{\ln|t|} \\
t\frac{dy}{dt} + t\frac{1}{t}y &= 2t \\
t\frac{dy}{dt} + y &= 2t \\
\frac{d}{dt}(ty) &= 2t \\
ty &= \int 2t \ dt \\
ty &= t^2 + C \\
y &= t + \frac{C}{t}
\end{aligned}$$

$y(1) = 3$ gives $C = 2$ and

$$y = t + \frac{2}{t}$$

(c) Trouble with this technique is two potentially challenging integrations:

- $\mu(t) = e^{-\int a(t) \ dt}$
- $\int b(t)\mu(t) \ dt$

Chapter 2: First order systems

1. We completed first order systems, now there are two possible next steps.

   (a) Single function y, higher order. Second order equations (think displacement, velocity, acceleration), then third and higher.

   (b) Multiple functions and first order systems of equations.

   We will choose the latter, though it turns out these are connected. Higher order linear equations can be rewritten as linear systems (and vice versa). We will return to high order equations in Chapter 3.

2. Again, three main approaches: This chapter takes the first two.

   (a) Qualitative: Trade direction fields for phase portraits which compare the derivatives of two unknown functions.

   (b) Numerical: Bend Euler's method to systems.

   (c) Analytic: Only special cases can be handled here. Chapter 3 focuses here.

.1 2.1: Modeling via systems

1. Example: Predator-prey first order system

$$\begin{cases} \frac{dR}{dt} &= 2R - 1.2F \cdot R \\ \frac{dF}{dt} &= -F + 0.9F \cdot R \end{cases}$$

where R is rabbit population and F is fox population. This is a nonlinear system since there are interaction terms ($F \cdot R$).

   (a) Equilibrium solutions: Require both $\dfrac{dR}{dt} = \dfrac{dF}{dt} = 0$.

$$\begin{aligned} R(2 - 1.2F) &= 0 \\ F(-1 + 0.9R) &= 0 \end{aligned}$$

   - If $R = 0$ and $F = 0$. So if no rabbits or fox, equilibrium.
   - If $R = \frac{1}{0.9} = \frac{10}{9}$, then need $F = \frac{2}{1.2} = \frac{20}{12} = \frac{5}{3}$. This is a nontrivial equilibrium where the system lies in balance.

   (b) Solution curves: Finding these explicitly is usually not possible and numerical methods are relied upon. Lucky for us this one twists into a separable equation.

$$\frac{dR/dt}{dF/dt} = \frac{dR}{dF} = \frac{R(2 - 1.2F)}{F(0.9R - 1)} \quad \Rightarrow \quad \int \frac{0.9R - 1}{R}\, dR = \int \frac{2 - 1.2F}{F}\, dF$$

   resulting in a family of solution curves

$$0.9R - \ln(R) = 2\ln(F) - 1.2F + C$$

   Plot a few in Desmos. Note we lose time (direction of the orbit). Euler's method brings back time in section 2.5.

   (c) Phase portrait: Plotting solution curves against R and F results in a phase portrait. Note our two equilibrium solutions.

   - $R = \frac{10}{9}, F = \frac{5}{3}$ is the center of the circle-like orbits. If start at that point, remain stationary.
   - $R = F = 0$ is also stationary. If only $F = 0$, $R \to \infty$. Also, if only $R = 0$, $F \to 0$ as expected with preyless foxes. To add direction to the phase portrait, just compute the tangents at any point on solution curves. If $R = F = 2$,

$$\frac{dR/dt}{dF/dt} = \frac{dR}{dF} = \frac{R(2 - 1.2F)}{F(0.9R - 1)} = \frac{2(2 - 1.2 \cdot 2)}{2(0.9 \cdot 2 - 1)} = \frac{2(20 - 12 \cdot 2)}{2(9 \cdot 2 - 10)} = \frac{-8}{16} = \frac{-1}{2}$$

     decreasing rabbits yet increasing foxes as we see in pplane.

   (d) This system is the famous Lotka-Volterra equations dating back to 1925.

2. Harmonic motion (mass on a spring)

   (a) Notation:

   - $y(t)$ = displacement ($y = 0$ at rest)
   - $y'(t)$ = velocity (speed and direction)
   - $y''(t)$ = acceleration ($y'' > 0$ speeding up, $y'' < 0$ slowing down)
   - Assuming no damping of spring, how do these relate?

   (b) Video demos:

   - Mass on spring: `https://www.youtube.com/watch?v=eeYRkW8V7Vg`
   - Paint can solution: `https://www.youtube.com/watch?v=p9uhmjbZn-c`

- Known as a harmonic oscillator (simple harmonic motion).

(c) Model formulation:

- Newton's second law: $F = ma$. For us,

$$F = my''$$

- Then how does the force of the spring F behave? Hook's law says

$$F = -ky.$$

Translation: Force is proportional to the stretching distance. More stretch, more force.

- Model:

$$F = ma \quad \Rightarrow \quad -ky = m\frac{d^2y}{dt^2} \quad \Rightarrow \quad m\frac{d^2y}{dt^2} + ky = 0$$

This is a second order equation. The solution should involve sine and cosine terms. If we added damping (friction on the spring), we would also include a $b\frac{dy}{dt}$ term to slow velocity (section 2.3).

(d) Solution: Use method of undetermined coefficients on $\cos(\omega t)$ (or $\sin(\omega t)$) to get general solution as

$$y = c_1 \cos\left(\sqrt{\frac{k}{m}}\right) + c_2 \sin\left(\sqrt{\frac{k}{m}}\right)$$

which can be combined into a single cosine curve as

$$y = R\cos(\omega t - \alpha)$$

with amplitude, frequency, and phase shift all depending on initial conditions $y(0)$ and $y'(0)$. More complete story on all this later.

(e) Can translate to a first order system for an alternate view. Let our two variables be y and $v = \frac{dy}{dt}$. Then, our second order equation becomes

$$\begin{cases} \frac{dy}{dt} = v \\ \frac{dv}{dt} = -\frac{k}{m}y \end{cases}$$

(f) Example: Simplify this by letting $\frac{k}{m} = 1$.

- Equilibrium solution: $y = v = 0$. No displacement or velocity.
- General solution is

$$y = c_1\cos(t) + c_2\sin(t)$$

If a specific solution is $y(0) = 1$, $y'(0) = 0$, then

$$y = \cos(t), \quad v = \sin(t)$$

and this curve is the unit circle in the $y - v$ plane.

- All solution curves are

$$y = R\cos(t - \alpha), \quad v = -R\sin(t - \alpha)$$

which are circles of radius R in the y, v plane.

(g) Second order equation vs first order system:

- Second order equation is simpler and directly translates mechanical law. Also good for finding analytic solution.
- System is more complex, but gives good qualitative analysis as well as better numeric solution.

.2 2.2: The geometry of systems

Jeff skip

1. Translating first order systems into the language of vectors makes it easier to see how phase portraits are generated.
   (a) Example: Predator prey
   $$\begin{cases} \frac{dR}{dt} &= 2R - 1.2F \cdot R \\ \frac{dF}{dt} &= -F + 0.9F \cdot R \end{cases}$$
   - Introduce vector $\vec{Y}$ and RHS vector field $\vec{F}(\vec{Y})$ as
   $$\vec{Y} = \langle R, F \rangle, \quad \vec{F}(\vec{Y}) = \langle 2R - 1.2F \cdot R, -F + 0.9F \cdot R \rangle$$
   - Then, at any point in the $R - F$ plane, we have a corresponding vector resulting from $\vec{F}$. If $R = F = 2$ as we saw last time,
   $$\vec{F} = \langle 4 - 4.8, -2 + 3.6 \rangle = \langle -0.8, 1.6 \rangle.$$
   - Plot all these vectors in the $R - F$ plane and we see our phase portrait.

   (b) Distinction: A direction field just gives direction while a phase portrait gives both magnitude and direction.
2. Homework: 1-18, 21

.3 2.3: The damped harmonic oscillator

Jeff skip

1. Here we accomplish two things:
   (a) Enhance the model for harmonic oscillation by adding damping (friction of the spring).
   (b) Introduce a general approach for second order constant coefficient equations:
   $$A\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Cy = 0$$
2. Damped harmonic oscillator:
   (a) We alter the precious model by adding damping in the form of $-b\frac{dy}{dt}$ for damping coefficient $b > 0$. Model: $F = ma$
   $$-ky - b\frac{dy}{dt} = m\frac{d^2y}{dt^2} \quad \Rightarrow \quad m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = 0$$
   Generally we rewrite to mimic quadratic equations (will see why shortly).
   $$A\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Cy = 0$$
   (b) Solution: It seems that the solution has to be an exponential $y = e^{st}$ for some number s. Plug it in and see.
   $$\begin{aligned} A\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Cy &= 0 \\ As^2e^{st} + Bse^{st} + Ce^{st} &= 0 \\ (As^2 + Bs + C)e^{st} &= 0 \\ As^2 + Bs + C &= 0 \end{aligned}$$

This quadratic is called the *characteristic equation* and gives a way to find s.

$$s = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

with three cases:

- $B^2 > 4AC$ (two real roots, over damping)
- $B^2 = 4AC$ (repeated real roots, critical damping)
- $B^2 < 4AC$ (complex roots, under damping)
- Also include $B = 0$ (no damping)

(c) Note this translates into a first order system: Dividing by A yields

$$\begin{cases} \frac{dy}{dt} = v \\ \frac{dv}{dt} = -\frac{C}{A}y - \frac{B}{A}v \end{cases}$$

and is of the form $\frac{d\vec{Y}}{dt} = \vec{F}(\vec{Y})$

(d) Example: $y'' + 3y' + 2y = 0$ has characteristic equation

$$s^2 + 3s + 2 = 0 \quad \Rightarrow \quad (s+2)(s+1) = 0 \quad \Rightarrow \quad s = -1, -2$$

Then the general solution is

$$y = c_1 e^{-t} + c_2 e^{-2t}.$$

This solution is stable and tends to zero. Can translate this solution to

$$\vec{Y} = c_1 \vec{Y_1} + c_2 \vec{Y_2}$$

for

$$\vec{Y_1} = \langle y_1, v_1 \rangle = \langle e^{-t}, -e^{-t} \rangle, \quad \vec{Y_2} = \langle y_2, v_2 \rangle = \langle e^{-2t}, -2e^{-2t} \rangle$$

and view in phase portrait.

(e) Can imagine interesting cases here:

- $s_1, s_2 > 0$ will be unstable (tend to infinity).
- $s_1 < 0, s_2 > 0$ will be unstable ans stable?
- $s_1 = s_2$ will take thinking to find a second solution.
- s complex will involve Euler's formula

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

resulting in sine and cosine terms (oscillation).

More on these later.

3. Homework: 1-10

.4 2.4: Additional analytic methods for special systems

Jeff skip

1. A general linear system is of the form

$$\begin{cases} \frac{dx}{dt} = Ax + By \\ \frac{dy}{dt} = Cx + Dy \end{cases}$$

For nice cases (decouple $B = C = 0$ or partially decoupled $B = 0$ or $C = 0$), we can draw on previous experience.

2. Example:

$$\begin{cases} \frac{dx}{dt} = 2x + 3y \\ \frac{dy}{dt} = -4y \end{cases}$$

   (a) Solve for y first:

$$y = c_1 e^{-4t}$$

   (b) Solve for x next:

$$\frac{dx}{dt} = 2x + 3c_1 e^{-4t}$$

   This is a nonhomogeneous equation which we can solve to get

$$x = c_2 x_h + x_p = c_2 e^{2t} - \frac{1}{2} c_1 e^{-4t}$$

3. Homework: 1-13

.5 2.5: Euler's method for systems

Jeff skip

1. Euler's method for first order equations extends to systems in a straightforward way.

   (a) First order equation: $\frac{dy}{dt} = f(t, y)$

$$y_{n+1} = y_n + \Delta t f_n$$

   Idea is to follow the tangent line by traveling stepsize Δt.

   (b) First order system:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

   Idea here is to follow the vector field by traveling $\Delta t \cdot \vec{F}(\vec{Y})$ where $\vec{F}$ is the vector at location (x, y).

$$\begin{aligned} \langle x_1, y_1 \rangle &= \langle x_0, y_0 \rangle + \Delta t \vec{F}(x_0, y_0) \\ \langle x_2, y_2 \rangle &= \langle x_1, y_1 \rangle + \Delta t \vec{F}(x_1, y_1) \\ &\vdots \\ \langle x_{n+1}, y_{n+1} \rangle &= \langle x_n, y_n \rangle + \Delta t \vec{F}(x_n, y_n) \end{aligned}$$

   which simply gives componentwise

$$x_{n+1} = x_n + \Delta t f_n, \quad y_{n+1} = y_n + \Delta t g_n.$$

2. Coding project assignment.

.6 2.6: Existence and uniqueness for systems

Jeff skip

1. Here we state an important theorem, no homework involved.

2. Theorem: Existence and uniqueness for systems.

   For $\vec{Y} = \langle x, y \rangle$ and $\vec{F} = \langle f(t, x, y), g(t, x, y) \rangle$, with $\vec{F}$ *continuously differentiable*, the system

   $$\frac{d\vec{Y}}{dt} = \vec{F}, \quad \vec{Y}(t_0) = \vec{Y}_0$$

   has a unique solution Y for t near t_0.

   Note, solutions may blow up in finite time as we saw in the one dimensional case.

3. Homework: None

.7 2.7: The SIR model of an epidemic

Jeff skip

1.

2. Coding assignment. Reproduce paper figures / tables.

.8 2.8: The Lorenz equations

Jeff skip

1.

2. Homework: None

Chapter 3: Linear systems

Chapter 2 introduced systems of DEs, though much of the time they cannot be solved. We are often limited to only qualitative (equilibrium solutions, nullclines, phase portraits) and quantititave analysis (Euler's method). One important case which we can solve is linear systems of the form

$$\frac{d\vec{y}}{dt} = A\vec{y}$$

where $\vec{y}$ is n-dimensions and A is a $(n \times n)$ constant coefficient matrix. This system can always be solved via linear algebra concepts. It is important for 2 reasons:

1. The systems appear in practice (damped harmonic oscillator).
2. Nonlinear systems can be approximated locally by linear systems which helps to classify their equilibrium point behavior (stable vs non-stable).

.1 3.1: Properties of linear systems and the linearity principle

1. Example: Formulate damped harmonic oscillator as a 2x2 linear system. Refresh matrix-vector multiplication.
2. General linear system, 2×2 case, $n \times n$ case.
3. Equilibrium points: Start with DHO example. General 2x2 case. Only one if A is non-singular (invertible). Invertible matrix theorem and check via determinant. 2x2 and 3x3 determinant eample. If singular, all of $Nul(A)$ is equilibrium points.

4. Theorem (EPs are easy): For non-singular matrix A, the only equilibrium points of $\frac{d\vec{y}}{dt} = A\vec{y}$ are at the origin.

5. Theorem (Linearity principle): For $\frac{d\vec{y}}{dt} = A\vec{y}$:

   - If $\vec{y}$ is a solution, then so is $c\vec{y}$.
   - If $\vec{y_1}$ and $\vec{y_2}$ are solutions, then so is $\vec{y_1} + \vec{y_2}$

   Proof follows from linearity of differentiation and linearity of matrix-vector multiplication. Combine the two into one as the superposition principle.

6. Theorem (General solutions): For $\vec{y_1}$ and $\vec{y_2}$ solutions to the linear system $\frac{d\vec{y}}{dt} = A\vec{y}$, if $\vec{y_1}(0)$ and $\vec{y_2}(0)$ are linearly independent, then we can always find a unique solution

$$\vec{y} = c_1\vec{y_1} + c_2\vec{y_2}$$

   to any initial value problem. $\vec{y} = c_1\vec{y_1} + c_2\vec{y_2}$ is known as the general solution to the system.

7. From the chapter 2 existence and uniqueness theorem, there is only one solution. This finds that unique solution. Our task is much narrower now: just find 2 particular solutions with LI initial values.

8. Example: SHO has 2 solutions $y_1(t) = \sin(t)$ and $y_2(t) = \cos(t)$. Reformulate as a linear system to get the general solution (the first component is the general solution to the 2nd order DE). Later we get a general approch which involves eigenstuff.

.2 old 3.1: Properties of linear systems and the linearity principle

1. Here we turn to analytic techniques for systems of equations, though we are tied to a special case with a wide range of applications: first order linear systems with constant coefficients:

   (a) 2×2 system:

$$\begin{cases} \frac{dx}{dt} = ax + by \\ \frac{dy}{dt} = cx + dy \end{cases}$$

   (b) Vector form:

$$\frac{d\vec{Y}}{dt} = A\vec{Y}$$

   where

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix}, \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

   (c) Recall matrix multiplication definition.

   (d) Note, results will easily extend to $n \times n$ systems.

   (e) Linearity principle (AKA superposition principle): If $\vec{Y_1}$ and $\vec{Y_2}$ solve the linear system, so do any linear combination $c_1\vec{Y_1} + c_2\vec{Y_2}$.

   Easy to show via derivative and matrix multiplication properties. Both are linear properties.

$$\frac{d}{dt}(c_1\vec{Y_1} + c_2\vec{Y_2}) = c_1\frac{\vec{Y_1}}{dt} + c_2\frac{\vec{Y_2}}{dt} = c_1A\vec{Y_1} + c_2A\vec{Y_2} = A(c_1\vec{Y_1} + c_2\vec{Y_2})$$

2. Recall high dimensional systems can be rewritten as linear systems. We will revisit the second order equations (harmonic oscillators) soon.

(a) Second order equation with damping:

$$m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = 0$$

(b) First order linear system: Substitute $v = \frac{dy}{dt}$ and name $\vec{Y} = \langle y, v\rangle$ giving

$$\frac{dY}{dt} = \begin{bmatrix} v \\ -\frac{k}{m}y - \frac{b}{m}v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}\vec{Y} = A\vec{Y}$$

3. For the general 2×2 system

$$\frac{d\vec{Y}}{dt} = A\vec{Y}$$

a key property is that matrix A is non-singular.

(a) Recall, A is nonsingular if

- $\det(A) \neq 0$, A^{-1} exists, only $\vec{0}$ in null(A), linearly independent columns, ...
- Refresh: `https://en.wikipedia.org/wiki/Invertible_matrix#The_invertible_matrix_theorem`

(b) Determinant of a 2×2 matrix A.

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

- Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- 3×3

(c) Theorem: For A nonsingular, the system $\frac{d\vec{Y}}{dt} = A\vec{Y}$ only equilibrium point is the origin $\vec{0}$.

4. General solutions to linear systems:

(a) Theorem: For $\vec{Y_1}, \vec{Y_2}$ solutions to the 2×2 linear system $\frac{d\vec{Y}}{dt} = A\vec{Y}$. If $\vec{Y_1}(0)$ and $\vec{Y_2}(0)$ are linearly independent, we can always find constants c_1 and c_2 such that

$$c_1\vec{Y_1} + c_2\vec{Y_2} \quad \text{(known as the general solution)}$$

solves the initial value problem.

(b) Definition: Vectors $\vec{x}$ and $\vec{y}$ are linearly independent if for any vector $\vec{a}$, there exists constants c_1 and c_2 such that

$$c_1\vec{x} + c_2\vec{y} = \vec{a}.$$

Note it is equivalent to say matrix $A = [\vec{x}\vec{y}]$ is nonsingular. Why?

(c) Example: Find the solution to the initial value problem

$$\frac{d\vec{Y}}{dt} = A\vec{A}, \quad A = \begin{bmatrix} 2 & 3 \\ 0 & -4 \end{bmatrix}, \quad \vec{Y}(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

We already found two solutions last time.

$$x = c_1e^{2t} - \frac{1}{2}c_2e^{-4t}, \quad y = c_2e^{-4t}.$$

Then two vector solutions are

$$\vec{Y_1} = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix}, \quad \vec{Y_2} = \begin{bmatrix} -\frac{1}{2}e^{-4t} \\ e^{-4t} \end{bmatrix}.$$

Then we need

$$c_1\vec{Y}_1(0) + c_2\vec{Y}_2(0) = \begin{bmatrix}1\\2\end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix}\vec{Y}_1(0) & \vec{Y}_2(0)\end{bmatrix}\vec{c} = \begin{bmatrix}1\\2\end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix}1 & -\frac{1}{2}\\0 & 1\end{bmatrix}\vec{c} = \begin{bmatrix}1\\2\end{bmatrix}$$

and we see why linear independence is again key. Result is $c_2 = 2$ and $c_1 = 2$ which leads to the solution to the IVP.

5. Homework: 5-19, 24, 30-32, 34-35

.3 3.2: Straight-line solutions

1. Find the general solution to $\frac{d\vec{y}}{dt} = A\vec{y}$ for

$$A = \begin{bmatrix}2 & 3\\0 & -4\end{bmatrix}$$

Think of all cases of limiting behavior. Plot a phase portrait of some solutions. Put one of these straight line solutions into the DE and see what A does.

2. Consider a general straight line solution, see the eigenstuff problem come out.

3. Carefully define eigenstuff. Note eigenvalues are unique, but eigenvectors are only unique up to scalar multiplication. Characteristic equation. Repeat and complex eigenvalues are possible. Mention diagonalization and decoupling.

4. Theorem: Eigenstuff gives a straight line solution. n distinct, real eigenvalues gives n linearly independent solutions and hence a full general solution.

5. Harmonic oscillator again.

.4 old3.2: Straight-line solutions

1. With this section we make an important connection to linear algebra (eigenvalues and eigenvectors).

2. Example: Same example as before.

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix}2 & 3\\0 & -4\end{bmatrix}\vec{Y} = A\vec{Y}$$

with general solution

$$\vec{Y} = c_1e^{2t}\begin{bmatrix}1\\0\end{bmatrix} + c_2e^{-4t}\begin{bmatrix}-\frac{1}{2}\\1\end{bmatrix} = c_1e^{2t}\vec{V}_1 + c_2e^{-4t}\vec{V}_2$$

   (a) pplane: Note straight lines in phase portrait coincide with two vectors. Also, direction followed relates to the sign of the exponent (decay or growth).

   (b) Once we are on a straight line, we remain on it. Reason:

$$A\vec{V}_1 = \begin{bmatrix}2 & 3\\0 & -4\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\0\end{bmatrix} = 2\vec{V}_1$$

3. $A\vec{V} = \lambda\vec{V}$ gives eigenvalues and eigenvectors (the characteristic direction of a matrix). Advantage of eigenness:

   (a) Matrix multiplication turns into scalar multiplication

   (b) Matrix diagonlization: If A has n linearly independent eigenvectors,

$$A = V\Lambda V^{-1} \quad \Rightarrow \quad A^k = V\Lambda^k V^{-1}$$

(c) How to find λ and $\vec{V}$?

- Eigenvalues: Need
$$A\vec{V} = \lambda\vec{V} \quad \Rightarrow \quad (A - \lambda I)\vec{V} = \vec{0}$$
for $\vec{V} \neq \vec{0}$. This implies $(A - \lambda I)$ is singular and
$$\det(A - \lambda I) = 0 \quad \text{(called the characteristic equation)}$$
- Eigenvectors: Once found λ, solve $(A - \lambda I)\vec{V} = \vec{0}$ via Gaussian elimination.

4. Example: Find the eigenvalues and eigenvectors of
$$A = \begin{bmatrix} -5 & -2 \\ -1 & -4 \end{bmatrix}$$

(a) Eigenvalues:
$$\det(A - \lambda I) = \begin{vmatrix} -5-\lambda & -2 \\ -1 & -4-\lambda \end{vmatrix} = (5+\lambda)(4+\lambda) - 2 = \lambda^2 + 9\lambda + 18 = (\lambda+6)(\lambda+3) = 0$$
Then $\lambda = -3, -6$.

(b) Eigenvectors:

- $\lambda = -3$
$$\begin{bmatrix} -5+3 & -2 \\ -1 & -4+3 \end{bmatrix} \vec{V} = \vec{0}$$
giving $\vec{V} = \langle 1, -1 \rangle$.
- $\lambda = -6$
$$\begin{bmatrix} -5+6 & -2 \\ -1 & -4+6 \end{bmatrix} \vec{V} = \vec{0}$$
giving $\vec{V} = \langle 2, 1 \rangle$.

5. Example: What can we expect from the linear system?
$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -5 & -2 \\ -1 & -4 \end{bmatrix} \vec{Y} = A\vec{Y}$$

(a) Straight line solutions are along eigenvectors. Both solutions decay along these lines.

(b) General solution is
$$\vec{Y} = c_1 e^{\lambda_1 t}\vec{V}_1 + c_2 e^{\lambda_2 t}\vec{V}_2 = c_1 e^{-3t}\langle 1, -1 \rangle + c_2 e^{-6t}\langle 2, 1 \rangle$$
Key here is *linearly independent eigenvectors* (as we saw in last section). Equivalently, need *distinct eigenvalues.*

(c) All solutions should decay.

6. Theorem: If matrix A has *distinct real* eigenvalues λ_k with corresponding eigenvectors $\vec{V}_k$, then
$$\frac{d\vec{Y}}{dt} = A\vec{Y}$$
has straight line solutions as
$$Y = c_k e^{\lambda_k t}\vec{V}_k$$
has general solution
$$Y = c_1 e^{\lambda_1 t}\vec{V}_1 + c_2 e^{\lambda_2 t}\vec{V}_2 + \cdots + c_n e^{\lambda_n t}\vec{V}_n$$

- Note dimension makes no difference here.
- Real distinct eigenvalues is important. We will look at complex and repeated eigenvalues later.

7. Homework: 1-25

.5 3.3: Phase portraits for linear systems with real eigenvalues

1. There are three main types of equilibrium points for 2×2 linear systems with real eigenvalues:
   (a) Sinks: $\lambda_1 < \lambda_2 < 0$ implies decay to $\vec{0}$ from all directions.
   (b) Sources: $0 < \lambda_1 < \lambda_2$ implies repelling from $\vec{0}$ to $\pm\infty$.
   (c) Saddles: $\lambda_1 < 0 < \lambda_2$ decays to $\vec{0}$ along a single straight line solution but repels from $\vec{0}$ to $\pm\infty$ otherwise.
2. Stability:
   (a) Sinks are stable meaning solutions near eachother stay near eachother. Small changes to the initial value result in small changes in the solution final value.
   (b) Sources and saddles are unstable meaning solutions near eachother can end up far away. Small changes to the initial value can result in largely different final values.
3. Examples:
   (a) Sink:
   $$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \vec{Y}$$
   has general solution
   $$\vec{Y} = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$
   (b) Source:
   $$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix} \vec{Y}$$
   has general solution
   $$\vec{Y} =$$
   (c) Saddle:
   $$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 5 & 4 \\ 9 & 0 \end{bmatrix} \vec{Y}$$
   has general solution
   $$\vec{Y} =$$
   (d) Plot phase portraits in pplane.
4. Homework: 1-16, 19-22, 27

.6 3.4: Complex eigenvalues

1. For 2×2 linear systems $\frac{d\vec{y}}{dt} = A\vec{y}$, the eigenstory has a few cases:
   (a) Two distinct real λ leading to linear independent eigenvectors.
   (b) Repeat real λ (next section)
   (c) $\lambda = 0$ (next section)
   (d) Complex λ, now and an impotant case.
2. Examples: $A = \begin{bmatrix} -2 & -3 \\ 3 & -2 \end{bmatrix}$ (textbook, sprial sink), $B = \begin{bmatrix} 2 & 1 \\ -4 & -1 \end{bmatrix}$ (textbook, sprial source)

3. Short review of complex numbers, arithmetic, plane, Euler's formula (capturing position and movement in the complex plane via a single exponential).

4. Writing down complex eigenstuff solution is easy, but how to make is useful in $\mathbb{R}^2$?

5. Theorem: For a complex valued solution $\vec{y}$ to the 2×2 linear system $\frac{d\vec{y}}{dt} = A\vec{y}$, we can decompose $\vec{y}$ as
$$\vec{y} = \vec{y}_{Re} + i\vec{y}_{Im}$$
and both $\vec{y}_{Re}$ and $\vec{y}_{Im}$ are solutions to the original system. Further, both are linearly independent when $t = 0$ giving general solution
$$\vec{y} = c_1\vec{y}_{Re} + c_2\vec{y}_{Im}.$$

6. Return to example and write out general solution.

7. Qualitative analysis: $\frac{d\vec{y}}{dt} = A\vec{y}$

   (a) For eigenstuff $\lambda = \alpha \pm \beta i$ and $\vec{v}, \overline{\vec{v}}$, we have complex solution via Euler's formula
   $$\vec{y} = e^{\alpha t}\left(\cos(\beta t) + i\sin(\beta t)\right)\vec{v}.$$

   (b) The sign of α gives exponental growth ($\alpha > 0$), decay ($\alpha < 0$), or pure oscillation ($\alpha = 0$).

   (c) β reveals perodicity for $0 \leq \beta t \leq 2\pi$.

   (d) Using the sum formulas for cosine and sine,
   $$\cos(u+v) = \cos(u)\cos(v) - \sin(u)\sin(v), \quad \sin(u+v) = \sin(u)\cos(v) + \cos(u)\sin(v),$$
   any linear combination of sine and cosine can be rewritten as a single shifted sinusoid:
   $$a\cos(\beta t) + b\sin(\beta t) = R\cos(\beta t - \gamma).$$
   This reveals periodic oscillation with period and a possible phase shift. Signal processing has a lot to say here.

   (e) Combining this gives 3 cases to draw: spiral sink, spiral source, or oscillation.

   (f) Note: The two components can have different amplitudes and phase shifts, which is why the resulting trajectory is generally an ellipse rather than a circle. Rotationalization (made up counterpart of diagonalization) reveals this.

.7 old3.4: Complex eigenvalues

1. There are four main cases with 2D linear systems:

   (a) Two distinct real eigenvalues

   (b) Repeated real eigenvalues (next section)

   (c) Zero eigenvalue (next section)

   (d) Complex eigenvalues (now)

   - We will see that complex eigenvalues result in rotations / oscillation. Hence the harmonic oscillator will return.

2. Example:
$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix}\vec{Y}, \quad \vec{Y}(0) = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

(a) Eigenvalues:

$$\det(A-\lambda I) = \begin{vmatrix} -2-\lambda & 1 \\ -1 & -2-\lambda \end{vmatrix} = (2+\lambda)^2 + 1 = 0$$

Then

$$\lambda_1 = -2+i, \quad \lambda_2 = -2-i.$$

(b) Eigenvectors: We need to solve the systems $A\vec{Y_i} = \lambda_i \vec{V_i}$ and the result is

$$\vec{V_1} = \begin{bmatrix} 1 \\ i \end{bmatrix}, \quad \vec{V_2} = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

(c) General solution: If we follow the mold from before, the result is

$$\vec{Y} = c_1 e^{\lambda_1 t}\vec{V_1} + c_2 e^{\lambda_2 t}\vec{V_2}$$

A few troubles here:

- How to make sense of $e^{(-2+i)t}$? Need Eulers formula:

  $$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

  Recall the power series formulation proof (`https://en.wikipedia.org/wiki/Euler%27s_formula`) and the use in the complex plane (unit circle). This formula is a big deal. Know it. Then,

  $$e^{(-2+i)t} = e^{-2t}e^{it} = e^{-2t}(\cos(t) + i\sin(t))$$

  Likewise for λ_2.
- We need $\vec{Y}$ to be real valued, not complex valued. Simplify to get use there.

$$\begin{aligned} \vec{Y_1} &= e^{(-2+i)t}\begin{bmatrix} 1 \\ i \end{bmatrix} \\ &= e^{-2t}(\cos(t) + i\sin(t))\begin{bmatrix} 1 \\ i \end{bmatrix} \\ &= e^{-2t}\begin{bmatrix} \cos(t) + i\sin(t) \\ i\cos(t) - \sin(t) \end{bmatrix} \\ &= e^{-2t}\begin{bmatrix} \cos(t) \\ -\sin(t) \end{bmatrix} + ie^{-2t}\begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} \end{aligned}$$

  $\vec{Y_2}$ results in the same two vectors. Can show these vectors both solve the system are linearly independent via

  $$\begin{vmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{vmatrix} = 1 \neq 0$$

Then the general solution is

$$\vec{Y} = c_1 e^{-2t}\begin{bmatrix} \cos(t) \\ -\sin(t) \end{bmatrix} + c_2 e^{-2t}\begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix}$$

Note no more complex numbers.

(d) IVP solution: $\vec{Y}(0) = \langle 6, 2\rangle$.

$$c_1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c1 \\ c2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

gives $c_1 = 6, c_2 = 2$ and

$$\vec{Y} = e^{-2t}\begin{bmatrix} 6\cos(t) + 2\sin(t) \\ -6\sin(t) + 2\cos(t) \end{bmatrix}$$

(e) Plot individual solutions to see oscillation:

$$x = e^{-2t}(\cos(t) + \sin(t)), \quad y = e^{-2t}(-\sin(t) + \cos(t))$$

Note e^{-2t} contributes decay, and the sine/cosine terms oscillation.

(f) Phase portrait: Note no straight line solutions, spiral sink in this case. Follow a single trajectory to see spiral.

3. Theorem: For $\vec{Y}$ a complex valued solution to $\dfrac{d\vec{Y}}{dt} = A\vec{Y}$ where

$$\vec{Y} = \vec{Y}_{re} + i\vec{Y}_{im},$$

then both $\vec{Y}_{re}$ and $\vec{Y}_{im}$ solve the system.

4. How do the eigenvalue real / imaginary parts contribute to the solution and overall behavior? Assume

$$\lambda = a \pm bi$$

(a) Sign of a assuming $b \neq 0$

- $a > 0$ equilibrium is spiral source
- $a < 0$ equilibrium is spiral sink
- $a = 0$ equilibrium is center (only circular motion)

(b) b cases

- $b > 0$ gives oscillation with frequency b
- $b = 0$ says real eigenvalues, no oscillation

(c) Rotation direction: At $\vec{Y} = \langle 1, 0\rangle$, compute $y' = c$.

- If $c > 0$, counterclockwise
- If $c < 0$, clockwise

5. Examples:

(a) Center:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \vec{Y}$$

- Eigenvalues: $\lambda = \pm 2i$
- Eigenvector:
- General solution:
- When is one cycle completed? $t = \pi$ periodic
- This is a circular orbit, but in general it can be an ellipse.

(b) Source:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 0 & 2 \\ -3 & 2 \end{bmatrix} \vec{Y}$$

(c) Eigenvalues: $\lambda = 1 \pm i\sqrt{5}$

(d) Eigenvector:

$$\vec{V} = e^{(1+\sqrt{5}i)t} \begin{bmatrix} 2 \\ 1 + i\sqrt{5} \end{bmatrix}$$

(e) General solution:

$$\vec{Y} = c_1 e^t \begin{bmatrix} 2\cos(\sqrt{5}t) \\ \cos(\sqrt{5}t) - \sqrt{5}\sin(\sqrt{5}t) \end{bmatrix} + c_2 e^t \begin{bmatrix} 2\sin(\sqrt{5}t) \\ \sqrt{5}\cos(\sqrt{5}t) + \sin(\sqrt{5}t) \end{bmatrix}$$

(f) Can still think about period and cycle as one revolution about the origin.

6. Stability:
   (a) Which of the above are stable?
   (b) Later we will go further to classify all stability cases of linear systems.
7. Homework: 1-24, 26

.8 3.5: Special cases: repeated and zero eigenvalues

1. Systems with only one eigenvalue or zero eigenvalue are relatively rare. We call them *degenerate* cases. They act as the "boundaries" between most common types of liner systems and play an important role for bifurcation analysis.
2. Repeat eigenvalue case: $\frac{d\vec{y}}{dt} = A\vec{y}$
   (a) Basic example: $A = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$
   (b) Motivating textbook example: $A = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$. Only one eigenvector, so no general solution. Have a look at the pplane.
   (c) Notice example is decoupled. Solve old way to fetch the general solution.
   (d) Note again, one repeat eigenvalue can lead to 2 LI eigenvectors, though not always.
   (e) Sound it all out: What if our general solution is of the form
   $$\vec{y} = e^{\lambda t}\vec{y}_0 + te^{\lambda t}\vec{v}$$
   where λ and $\vec{v}$ are eigenstuff as above. Then the solution works so long as $\vec{v}$ is an eigenvector and
   $$\vec{v} = (A - \lambda I)\vec{y}_0.$$
   (f) Theorem: Suppose $\frac{d\vec{y}}{dt} = A\vec{y}$ and A has a repeat eigenvalue λ with only one eigenvector $\vec{v}$, then the general solution is
   $$\vec{y} = e^{\lambda t}\vec{y}_0 + te^{\lambda t}\vec{v}$$
   where
   $$\vec{v} = (A - \lambda I)\vec{y}_0.$$
   Note, if it turns out $\vec{v} = \vec{0}$, then the initial condition is an eigenvector.

.9 old3.5: Special cases: repeated and zero eigenvalues

1. Two special cases remain (degenerate systems). These play an importation role when studying system bifurcations.
   (a) Repeated real eigenvalues (may still have 2 linearly independent eigenvectors, or maybe only 1). Note repeat complex eigenvalues not possible since one complex eigenvalue implies there is a conjugate pair.
   (b) Zero eigenvalue (whole line of equilibrium solutions)
2. Repeated eigenvalue case
   (a) Complete case: still 2 linearly independent eigenvectors (actually infinitely many)
   $$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \vec{Y}$$

- Note repeated eigenvalue are $\lambda = 2$ and can see that

$$\vec{V_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{V_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

  are both eigenvectors (though there are many more).
- General solution:

$$\vec{Y} = c_1 e^{\lambda t}\vec{V_1} + c_2 e^{\lambda t}\vec{V_2} = e^{\lambda t}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

  and there are infinite straight line solutions.
- Pplane visual.
- The sign of λ determines source or sink. No spiraling possible here.

(b) Incomplete case: only 1 linearly independent eigenvector

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}\vec{Y}$$

- Eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} -2-\lambda & 1 \\ 0 & -2-\lambda \end{vmatrix} = (\lambda + 2)^2 = 0$$

  gives $\lambda = -2$.
- Eigenvector: $A\vec{V} = \vec{V}$ yields only

$$\vec{V} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

- Single solution: $Y = c_1 e^{-2t}\langle 1, 0\rangle$, but we know there should be two linearly independent solutions.
- Switch gears and solve as a (partially) decoupled system.

$$\frac{dx}{dt} = -2x + y, \quad \frac{dy}{dt} = -2y$$

  Result is

$$y = y_0 e^{-2t}, \quad x = y_0 t e^{-2t} x_0 e^{-2t}$$

  giving our system general solution of

$$\vec{Y} = e^{-2t}\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + te^{-2t}\begin{bmatrix} y_0 \\ 0 \end{bmatrix} = e^{-2t}\vec{Y_0} + y_0 te^{-2t}\vec{V} = e^{-2t}\vec{Y_0} + te^{-2t}(A - \lambda I)\vec{Y_0}$$

- This final solution form can be checked in general:

$$\vec{Y} = e^{\lambda t}\vec{Y_0} + te^{\lambda t}(A - \lambda I)\vec{Y_0}$$

  solves $\frac{d\vec{Y}}{dt} = A\vec{Y}$, and hence is the general solution. Plug into both sides and see.

(c) Theorem: For linear system

$$\frac{d\vec{Y}}{dt} = A\vec{Y}$$

with repeated eigenvalue λ and only one linearly independent eigenvectors, the general solution is

$$\vec{Y} = e^{\lambda t}\vec{Y_0} + te^{\lambda t}\vec{V}$$

where

$$\vec{V} = (A - \lambda I)\vec{Y_0}.$$

- If $\vec{V} = \vec{0}$, then there are only straight line solutions as we saw in the first example.
- If $\vec{V} \neq \vec{0}$, then it is an eigenvector as in the second example.
- This theorem should be used from the start to solve any system with repeated eigenvalues. Our conversation stumbled across it.

(d) Example:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -2 & 1 \\ -1 & 0 \end{bmatrix} \vec{Y}$$

has repeated eigenvalue $\lambda = -1$. Then the general solution is

$$\vec{Y} = e^{-t} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + te^{-t}(A + I) \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = e^{-t} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + te^{-t} \begin{bmatrix} -x_0 + y_0 \\ -x_0 + y_0 \end{bmatrix}$$

Check that it works to be sure.

3. Zero eigenvalue case: This case is simpler. One example will suffice.

(a) Example:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \vec{Y}$$

- Eigenvalues:

$$\det(A - \lambda I) = (4 - \lambda)(1 - \lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0$$

$\lambda = 0, 5$. Note that implies $\det(A) = 0$ and this matrix is nonsingular and the nullspace has more than $\vec{0}$. Our eigenvector should live there too.

- Eigenvectors: $A\vec{V_1} = 5\vec{V_1}$ gives

$$\vec{V_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

while $A\vec{V_2} = \vec{0}$ gives

$$\vec{V_2} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

- General solution:

$$\vec{Y} = c_1 e^{5t} \vec{V_1} + c_2 \vec{V_2}$$

- Every point along $\vec{V_2}$ will serve as an equilibrium point (unstable in this case, stable for $\lambda < 0$). All other solutions will follow lines parallel to $\vec{V_1}$.

4. Homework: 1-23

.10 3.6: Second-order linear equations

1. Here we give the complete unified story of the second-order linear equations:

- Second order equation:

$$m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = 0 \quad \Rightarrow \quad \frac{d^2y}{dt^2} + p\frac{dy}{dt} + qy = 0$$

where constants denote m (mass), b (damping constant), and k (spring strength). Engineer a leading coefficient of 1 for simplicity.

- Equivalent first order linear system:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 0 & 1 \\ -q & -p \end{bmatrix}$$

where $\vec{Y} = \langle y, \frac{dy}{dt} \rangle = \langle y, v \rangle$.

- It will turn out our characteristic equation from chapter 2 matches our eigenvalue equation from this chapter.

2. Example:
$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0$$
   (a) Characteristic equation approach: Assume $y = e^{st}$, then the differential equation reduces to
$$e^{st}(s^2 + 3s + 2) = 0 \quad \Rightarrow \quad s^2 + 3s + 2 = (s+2)(s+1) = 0 \quad \Rightarrow \quad s = -2, -1$$
   Then the general solution is
$$y = c_1 e^{-2t} + c_2 e^{-t}.$$
   Note, the solution tends to zero without oscillation.
   (b) Plot the solution y in Desmos for $y(0) = 1$, $y'(0) = v(0) = 1$ giving $c_1 = 2, c_2 = -1$.
   (c) Linear system / eigenvalue approach: Let $\vec{Y} = \langle y, v \rangle$. Then,
$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \vec{Y}$$
   . Then,
$$\det(A - \lambda I) = -\lambda(-3 - \lambda) + 2 = \lambda^2 + 3\lambda + 2 = 0$$
   the same as the characteristic equation. $\lambda = -2, -3$ and the general solution is
$$\vec{Y} = c_1 e^{-3t}\vec{V}_1 + c_2 e^{-2t}\vec{V}_2.$$
   where eigenvectors are found to be
$$\vec{V}_1 = \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}, \quad \vec{V}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$
   and we have the same y as found above.
   (d) Graph in pplane. Note equilibrium is a sink (stable).

3. Example:
$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 13y = 0, \quad y(0) = 1, y'(0) = -4$$
   (a) General solution: Characteristic equation is
$$s^2 - 4s + 13 = 0 \quad \Rightarrow \quad s = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$$
   giving
$$e^{(-2+3i)t} = e^{-2t}(\cos(3t) + i\sin(3t))$$
   for the general solution
$$y = c_1 e^{-2t}\cos(3t) + c_2 e^{-2t}\sin(3t).$$
   (b) Initial condition gives $c_1 = 1, c_2 = -2$ with solution
$$y = e^{-2t}\cos(3t) - 2e^{-2t}\sin(3t).$$
   Graph the result in Desmos.
   (c) Give phase portrait for equivalent system
$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 0 & 1 \\ -13 & 4 \end{bmatrix} \vec{Y}$$

(d) Can see the strength provided by a and b in the eigenvalue $a + bi$. This can be generalized in terms of m (mass), b (damping constant), and k (spring strength)

4. Summary of general harmonic oscillator cases:

$$m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = 0 \quad \Rightarrow \quad ms^2 + bs + k = 0 \; quad \Rightarrow \quad s = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

(a) $b = 0$ gives no damping, only oscillation due to purely imaginary eigenvalues.

$$\cos(\omega t), \sin(\omega t)$$

(b) $b^2 > 4mk$ gives overdamping, only exponential decay.

$$e^{s_1 t}, e^{s_2 t}$$

(c) $b^2 = 4mk$ gives critical damping, only exponential decay due to repeated root.

$$e^{s_1 t}, te^{s_1 t}$$

(d) $b^2 < 4mk$ gives underdamping, decay plus oscillation

$$e^{at}\cos(\omega t), e^{at}\sin(\omega t)$$

(e) Classify the above two examples via this simple test. Have seen many examples in past. More in the homework. Sample solutions for 3 cases:

- No damping
$$\frac{d^2y}{dt^2} + 0\frac{dy}{dt} + y = 0, \quad s = \pm i$$
- Overdamping
$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + y = 0, \quad s = -\frac{3}{2} \pm \frac{\sqrt{5}}{2}$$
- Critical damping
$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0, \quad s = -1$$
- Underdamping
$$\frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0, \quad s = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

5. Homework: 1-33

.11 3.7: The trace-determinant plane

1. In the last section we unified the behavior of all harmonic oscillators. Now we tackle all first order linear systems.

$$\frac{d\vec{Y}}{dt} = A\vec{y} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \vec{Y}$$

with $\vec{Y} = \langle x, y \rangle$. Given A, we will answer two main questions.

(a) Is the system stable? That is, if two solutions start close together, will they remain close together?

(b) What is the global behavior of the system? Cases are

- Sink, saddle, source, spirals, etc.

(c) Both of these questions are determined by the eigenvalues of A, but a simpler approach is to consider the trace and determinant of A.

$$\text{tr}(A) = a + d, \quad \det(A) = ad - bc$$

2. The trace-determinant plane:

(a) Eigenvalues from trace / determinant:

$$\det(A-\lambda I) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda)-bc = \lambda^2-(a+d)\lambda+(ad-bc) = \lambda^2-\text{tr}(A)\lambda+\det(A) = 0$$

So the trace and determinant determine λ via

$$\lambda = \frac{\text{tr}(A) \pm \sqrt{\text{tr}(A)^2 - 4\det(A)}}{2}$$

(b) Real / complex cases:

- $\text{tr}(A)^2 > 4\det(A)$ gives two distinct real eigenvalues
- $\text{tr}(A)^2 < 4\det(A)$ gives complex eigenvalues
- $\text{tr}(A)^2 = 4\det(A)$ gives repeated real eigenvalues, barrier between real and complex cases.

(c) Activity: Quiz bonus points. Give the trace / determinant plane with phase portraits covered up. Have them fill in the behavior of each then uncover. Options:

- Sink, degenerate sink, spiral sink, saddle, line of stable fixed points, line of unstable fixed points, center, spiral source, degenerate source, source.

Which ones are stable?

(d) Stability cases:

- Note also that $\det(A) = \lambda_1 \cdot \lambda_2$ because of the factoring of the characteristic equation. If real eigenvalues only, need $\text{tr}(A) < 0$(at least one negative eigenvalue) and $\det(A) > 0$ (product of eigenvalues same sign). If complex eigenvalues need $\text{tr}(A) < 0$.
- Stable when $\text{tr}(A) < 0$ and $\det(A) > 0$.
- Unstable otherwise.

(e) This conversation is especially useful for bifurcation analysis which you will focus on in the homework.

3. Examples: Determine behavior and stability of each via the trace and determinant.

(a) $A = \begin{bmatrix} 0 & -1 \\ 2 & -3 \end{bmatrix}$ is stable.

(b) $A = \begin{bmatrix} 1 & -1 \\ 3 & 3 \end{bmatrix}$ is unstable.

(c) $A = \begin{bmatrix} 0 & 4 \\ 5 & -6 \end{bmatrix}$ is unstable.

(d) $A = \begin{bmatrix} 0 & -7 \\ 7 & 0 \end{bmatrix}$ is neutral.

4. Final visual: `https://i.stack.imgur.com/duPPi.png`

5. Homework: 1-12

.12 3.8: Linear systems in three dimensions

Jeff skip

1. What remains of the course?
   (a) What have we done?
   - First order equations, second order linear equations, 2×2 linear systems

   (b) What remains?
   - High order linear systems, $n \times n$ linear systems, of course these are tied together. I will only give basic idea, no homework.
   - Nonlinear systems, we will borrow techniques from linear systems to partially analyze.
   - New idea all together: Laplace transform, change our space from derivatives / functions to algebra and back.

2. High order linear equations: Consider third order, but ideas extend higher.

$$y''' + By'' + Cy' + Dy = 0$$

   (a) Transfer to a 3×3 linear system. Denote

$$\vec{Y} = \begin{bmatrix} y \\ y' \\ y'' \end{bmatrix}$$

   then we are left with the system

$$\frac{d\vec{Y}}{dt} = A\vec{Y} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -D & -C & -B \end{bmatrix} \vec{Y}$$

   (b) Eigenvalues: Characteristic equation again.

$$\det(A - \lambda I) = -(\lambda^3 + B\lambda^2 + C\lambda + D) = 0$$

   Challenge here is cubic equations are harder to deal with.

   (c) Eigenvectors: For this case, eigenvectors have the special form

$$\vec{V} = \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \end{bmatrix}$$

   Multiply $A\vec{V}$ and use the characteristic equation to see.

   (d) High dimensional systems are more difficult to analyze, though familiar features appear (sink, source, saddle, rotation, etc). See text for concrete examples.

3. Homework: None. Not on quiz / exam.

Chapter 4: Forcing and resonance

Jeff skip all chapter

.1 4.1: Forced harmonic oscillators

.2 4.2: Sinusoidal forcing

.3 4.3: Undamped forcing and resonance

.4 4.4: Amplitude and phase of the steady state

.5 4.5: The Tacoma Narrows Bridge

Chapter 5: Nonlinear systems

Here we touch on 2×2 nonlinear, autonomous systems of the form

$$\frac{dx}{dt} = f(x, y), \quad \frac{dy}{dt} = g(x, y)$$

where f and g are nonlinear functions of x and/or y. Of course analysis will be harder here, but we can borrow ideas of linear systems to do analysis at specific points (equilibrium points) / curves (isoclines).

.1 5.1: Equilibrium point analysis

1. Motivating example:

$$\frac{dx}{dt} = -2x + 2x^2 = f(x, y), \quad \frac{dy}{dt} = -3x + y + 3x^2 = g(x, y)$$

   (a) Phase portrait is complicated. Let's look closer.

   (b) Equilibrium points:

$$f(x, y) = 2x(-1 + x) = 0, \quad g(x, y) = -3x + y + 3x^2 = 0$$

   Two points work: $(0, 0), (1, 0)$. Zoom into these on phase portrait to see a saddle point and a source.

   (c) Idea of linearization: Replace nonlinear system with a linear one near equilibrium points.

2. Linearization of a nonlinear equation: The logistic equation.

$$\frac{dy}{dt} = y(1 - y) = f(y)$$

   (a) Equilibrium points: $f(y) = 0$ at $y = 0, 1$.

   (b) Stability: Draw the phase line to see stability (sink or source). Sketch direction field resulting from the phase portrait. Comparing to the graph of f, we just care if f is increasing or decreasing thru the equilbrium point.

   (c) Linearization approach to stability:

   - Consider $y = 1$. Approximate f by the tangent line thru this equilibrium point. Taylor series approximation gets us there.

$$f(y) = f(1) + f'(1)(y - 1) + \frac{f''(1)}{2!}(y - 1)^2 + \cdots$$
$$f(y) \approx f(1) + f'(1)(y - 1) = 0 + f'(1)(y - 1)$$

   - Linearized equation centered at $y = 1$.

$$\frac{d(y - 1)}{dt} = f'(1)(y - 1) = -(y - 1)$$

     Note we shifted to be centered at zero to make easier to solve. Solution:

$$y - 1 = Ce^{-t}$$

     This solution converges to zeros on the RHS and hence is stable.

- Linearized equation centered at $y = 0$.

$$\frac{d(y-0)}{dt} = f'(0)(y-0) = (y-0) \quad \Rightarrow \quad y = Ce^t$$

which is unstable (grows to ∞).

(d) General linearization of a single nonlinear equation centered at Y:

$$\frac{dy}{dt} = f(y) \quad \Rightarrow \quad \frac{d(y-Y)}{dt} = f'(Y)(y-Y)$$

Replace f with a linear approximation centered at the equilibrium. It is an approximation, though exact at the equilibrium which is our focus here. Here, the sign of f' at equilibrium determines stability ($f' > 0$ unstable, $f' < 0$ stable).

(e) This idea extends to 2 dimensions where nonlinear life is much harder otherwise.

3. Linearization of nonlinear system:

$$\frac{dx}{dt} = -2x + 2x^2 = f(x,y), \quad \frac{dy}{dt} = -3x + y + 3x^2 = g(x,y)$$

(a) Equilibrium points: $(x,y) = (0,0), (1,0)$.

(b) Linearization centered at $(1,0)$: Replace f (and g) with a linear function centered at $(1,0)$.

$$f(x,y) = f(1,0) + \frac{\partial f}{\partial x}(x-1) + \frac{\partial f}{\partial y}(y-0) + \cdots$$
$$f(x,y) \approx f(1,0) + \frac{\partial f}{\partial x}(x-1) + \frac{\partial f}{\partial y}(y-0) = \frac{\partial f}{\partial x}(x-1) + \frac{\partial f}{\partial y}(y-0)$$

Then, the linearized equation for x becomes

$$\frac{d(x-1)}{dt} = \frac{\partial f}{\partial x}(x-1) + \frac{\partial f}{\partial y}(y-0).$$

Likewise, for g we have

$$\frac{d(y-0)}{dt} = \frac{\partial f}{\partial x}(x-1) + \frac{\partial f}{\partial y}(y-0).$$

Combining the two, we have the system

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x}(1,0) & \frac{\partial f}{\partial y}(1,0) \\ \frac{\partial g}{\partial x}(1,0) & \frac{\partial g}{\partial y}(1,0) \end{bmatrix} \vec{Y} = A\vec{Y}$$

where

$$\vec{Y} = \begin{bmatrix} x-1 \\ y-0 \end{bmatrix}$$

Note the shifting to center at $(1,0)$ was necessary to get our linear system matching on both sides.

(c) Definition: The Jacobian matrix is

$$A = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix}$$

(d) Stability analysis at $(1,0)$: Differentiating, our system becomes

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \vec{Y} = A\vec{Y}$$

Take our trace-determinant plane analysis from last chapter.

- $\text{tr}(A) = 3 > 0$ and $\det(A) = 2 > 0$ so we are unstable.
- $\text{tr}(A)^2 > 4\det(A)$, so this is a source.

Alternatively eigenvalues / vectors can be found.

- Eigenvalues: 2,1
- Eigenvectors: $\lambda = 2$ gives $\langle 1, 3\rangle$ and $\lambda = 1$ gives $\langle 0, 1\rangle$.

(e) Repeat at second equilibrium point $(0, 0)$.

- Linearization:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} -2 & 0 \\ -3 & 1 \end{bmatrix} \vec{Y} = A\vec{Y}$$

- Stability: $\text{tr}(A) = -1 < 0, \det(A) = -2 < 0$ so unstable and also a saddle:
- Eigenvalues are $\lambda = -2, 1$ with respective eigenvectors $\langle 1, 1\rangle$, $\langle 0, 1\rangle$.

(f) Bring back the phase portrait.

4. Example: Competing species model:

$$\frac{dx}{dt} = 2x(1 - x/2) - xy, \quad \frac{dy}{dt} = 3y(1 - y/3) - 2xy$$

(a) These are two separate logistic equations with added xy interaction terms. For the x equation, if y gets much larger than x, the population of x will decrease due to increased competition.

(b) Equilibriums:

- $\frac{dx}{dt} = 0$ when $x = 0$. This gives $y = 0, 3$ in equation 2.
- $\frac{dy}{dt} = 0$ when $y = 0$. This gives $x = 0, 2$ in equation 1.
- Note these are all points on the x and y axis. There must be solutions in the first quadrant (other quadrants make no sense) because of the existence / uniquiness theorem.

$$\begin{cases} x(2 - x - y) = 0 \\ y(3 - y - 2x) = 0 \end{cases}$$

Assuming $x, y \neq 0$,

$$\begin{cases} 2 - x - y = 0 \\ 3 - y - 2x = 0 \end{cases}$$

leading to $x = y = 1$.

(c) $(1, 1)$ is the only equilibrium allowing for coexistence. Let's see its behavior.

- Linearization at $(1, 1)$:

$$\frac{d\vec{Y}}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} \vec{Y} = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix} \vec{Y}$$

where $\vec{Y} = \langle x - 1, y - 1\rangle$.

- Eigenvalues: $\lambda = -1 \pm \sqrt{2}$ leads to a saddle point (unstable). Sad fate for one of the two species. Note on eigenvector divides the phase plane to decide which species will survive long term.

(d) Phase portrait.

5. Homework: 1-16

.2 5.2: Qualitative analysis

1. Nonlinear system:
$$\frac{dx}{dt} = f(x,y), \quad \frac{dy}{dt} = g(x,y)$$
   (a) Linearization is only accurate near the equilibrium point (both $f = 0$ and $g = 0$) for which it is centered. Everything else in the phase portrait is a mystery.
   (b) Idea of nullclines: Consider $f = 0$ ($x-$nullcline) separate from $g = 0$ (y-nullcline). This will allow us to sharpen our knowledge of solution flow throughout the phase portrait.

2. Example: Competing species model (in first quadrant only)
$$\frac{dx}{dt} = f(x,y) = 2x(1 - x/2) - xy, \quad \frac{dy}{dt} = g(x,y) = 3y(1 - y/3) - 2xy$$
   (a) $x-$nullcline: Set $\frac{dx}{dt} = f(x,y) = 0$.
$$2x(1 - x/2) - xy = x(2 - x - y) = 0 \text{ gives } x = 0, y = -x + 2$$
   These two curves are the $x-$nullclines.
   (b) $y-$nullcline: Set $\frac{dy}{dt} = g(x,y) = 0$.
$$3y(1 - y/3) - 2xy = y(3 - y - 2x) = 0 \text{ gives } y = 0, y = -2x + 3$$
   These two curves are the $y-$ nullclines.
   (c) Graph all 4 nullclines in the $xy-$plane.
   - Note, $x-$nullclines only allow change in y hence the direction vector field has to be vertical. Plug a point into $\frac{dy}{dt} = g(x,y)$ to see if point up (positive slope) or down (negative slope). Up/down direction cannot change unless cross a $y-$nullcline.
   - Likewise for $y-$nullclines.
   - Intersections of these 4 nullclines gives equilibrium points.
   - Note, these nullclines are not eigenvectors and in general will not be straight lines. They are any curve decided by f and g.

   (d) Nullclines divide the first quadrant into 4 regions (A, B, C, D).
   - Let A be the top left region. $x = 0$ is a solution curve. Also, boundaries of A point into A (up and left). Hence any solution starting in A has to stay in A trending up and right to the equilibrium point $(0, 3)$. Note, existence and uniqueness theorem says
   - Similar for region B (lower right).
   - Region C (lower left) is bordered below by solution curves and has three options: enter A, enter B, or approach equilibrium $(1, 1)$ via saddle point eigenvector.
   - Region D is similar to C though from above.

3. Example: Graph the nullclines and complete phase portrait in the first quadrant.
$$\frac{dx}{dt} = x(-x - y + 40), \quad \frac{dy}{dt} = y(-x^2 - y^2 + 2500)$$
   (a) Nullclines:
   - $\frac{dx}{dt} = 0$ when $x = 0$ or $y = -x + 40$.
   - $\frac{dy}{dt} = 0$ when $y = 0$ or $x^2 + y^2 = 50^2$.
   - Find the horizontal / vertical vector field for each.

   (b) Phase portrait:
   - Equilibrium: $(0, 0)$ is a source, $(0, 50)$ is a sink, $(40, 0)$ is a saddle.
   - Sketch and check in pplane.
   - It is clear here that nullclines really give the form of the phase portrait.

4. Homework: 1-14

.3 5.3: Hamiltonian systems

Jeff skip

.4 5.4: Dissipative systems

Jeff skip

.5 5.5: Nonlinear systems in three dimensions

Jeff skip

.6 5.6: Periodic forcing of nonlinear systems and chaos

Jeff skip

Chapter 6: Laplace transforms

.1 6.1: Laplace transforms

1. Here we see a new idea: Laplace transform
   (a) Idea:
   - Transform a differential equation into an algebraic equation ($y'(t)$ in time domain $\to$ $sY(s)$ in frequency domain).
   - Solve algebra equation (for $Y(s)$ in frequency domain).
   - Transform back (to $y(t)$ in time domain).

   (b) Advantage of Laplace transform:
   - Process to make differential equations easier, though inverting the transformation is often the hardest part.
   - Can handle functions we couldn't before (step function (Heaviside function, turn on a signal switch), impulse function (delta function, sudden impulse of explosion or laser)).
2. Laplace transform intution: There are two sources of origin for Laplace trasform.
   (a) Mathematical: Generalized idea of a power series.
   - Given a sequence $\{a_n\}$, power series is of the form
   $$\sum_{n=1}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots = f(x)$$
   which may converge to function f for certain x (interval of convergence).
   - Examples:
   $$\sum x^n = \frac{1}{1-x}, |x| < 1, \quad \sum \frac{1}{n!} x^n = e^x, \text{ for all } x, \quad \text{etc}$$
   - Think of a power series as a transformation from discrete sequence a_n to function $f(x)$. Depending on a_n, this transformation may only make sense for some x values. Also, a_n and f have different meaning and usefulness.
   - Laplace transform is the same transformation, only for continuous functions $a(t)$ (in place of a_n) to new functions $F(x)$ (we use s to denote signal space as seen below). Our summation becomes an improper integral then.
   $$F(x) = \int_0^{\infty} a(t) x^t \, dx$$

Base x is not so nice here and it is good to change to base e instead.

$$F(x) = \int_0^\infty a(t)e^{\ln(x)t}\ dx$$

Since we need $0 < x < 1$ for convergence, we will have $\ln(x) < 0$. Do a change of variable naming $-s = \ln(x)$ and we get

$$F(s) = \int_0^\infty a(t)e^{-st}\ ds.$$

- The above line is the definition of the Laplace transform. Note that $a(t)$ cannot grow faster than an exponential otherwise we have no hope for convergence.
- Further, thru Euler's formula

$$e^{(ta+bi)t} = e^t(\cos(bt) + i\sin(bt))$$

makes base e ideal for complex numbers and handling signals over time.

(b) Signal processing / control theory:

- Signal processing has a wide range of application (audio, circuits, light, radar, telecommunications, etc) and took off during the tech boom of world war 2.
- Video intuition for Fourier transform (a special case of Laplace transform): `https://www.youtube.com/watch?v=spUNpyF58BY`

3. Definition of Laplace transform:

(a) The Laplace transform $Y(s)$ of function $y(t)$ is defined as

$$Y(s) = \mathcal{L}[y(t)] = \int_0^\infty y(t)e^{-st}\ dt$$

with domain all s for which the improper integral converges. In general s can be a complex number via Euler's formula.

(b) Terminology note. A transform differs from an operator. Operators take objects and map them to objects in a similar space (eg differential operator). Transformations change the space (in our case time t to signal s).

4. Examples: This transform is easy to compute. Just an improper integral from calculus.

(a) $y(t) = 1$ (akin to geometric series above)

$$\begin{aligned} Y(s) = \mathcal{L}[1] &= \int_0^\infty 1e^{-st}\ dt \\ &= \lim_{b\to\infty} \int_0^b e^{-st}\ dt \\ &= \lim_{b\to\infty} \frac{1}{-s}e^{-st}\Big|_0^b \\ &= \lim_{b\to\infty} \frac{1}{-s}(e^{-sb} - 1) \\ &= \frac{1}{s} \end{aligned}$$

Note that we need $s > 0$ for this to work. Make sure to keep track of the domains of the Laplace transforms computed.

(b) $y(t) = e^{at}$ for some constant a since these are our prime targets for this course. Two ways to compute.

- Use formula as above:

$$Y(s) = \mathcal{L}[e^{at}] = \int_0^\infty e^{at}e^{-st}\ dt = \int_0^\infty e^{(a-s)t}\ dt = \cdots$$

  The book does this.
- Use $\mathcal{L}[1]$ result directly. For any function $f(t)$,

$$\mathcal{L}[e^{at}f(t)] = \int_0^\infty f(t)e^{-(s-a)t}\ dt = F(s-a)$$

  Then, since $\mathcal{L}[1] = \frac{1}{s}, s > 0$, we have that

$$\mathcal{L}[e^{at} \cdot 1] = \frac{1}{s-a}, s > a.$$

  Multiplication by e^{at} inside the Laplace transform works as a horizontal shift.

(c) Will channel Euler's formula and the use the transform for $\mathcal{L}[e^{at}]$ to derive formulas for sine and cosine. Try on own if ambitious.

5. Laplace transform advantage with differential equations: Derivatives become powers of s and $\mathcal{L}$ is a linear transformation.

(a) Theorem: Laplace transform of derivatives.

$$\mathcal{L}\left[\frac{dy}{dt}\right] = s\mathcal{L}[y] - y(0)$$

Proof: This is just integration by parts in disguise.

$$\begin{aligned}\mathcal{L}\left[\frac{dy}{dt}\right] &= \int_0^\infty \frac{dy}{dt}e^{-st}\ dt \\ &= \lim_{b\to\infty}\left(ye^{-st}\Big|_0^b - \int_0^b -sye^{-st}\ dt\right) \\ &= s\mathcal{L}[y] - y(0)\end{aligned}$$

Here we need to assume again that y has no more than exponential growth in order for ye^{-st} to converge to zero.

(b) Example: Can iterate for high order derivatives.

$$\mathcal{L}\left[\frac{d^2y}{dt^2}\right] = s\mathcal{L}\left[\frac{dy}{dt}\right] - y'(0) = s\left(s\mathcal{L}[y] - y(0)\right) - y'(0) = s^2\mathcal{L}[y] - sy(0) - y'(0)$$

(c) Example: Can now unravel powers of t: t^n.

$$\mathcal{L}[t] = \frac{1}{s}\left(\mathcal{L}[1] + y(0)\right) = \frac{1}{s^2}$$

$$\mathcal{L}[t^2] = \frac{1}{s}\left(\mathcal{L}[2t] + y(0)\right) = \frac{2}{s^3}$$

$$\mathcal{L}[t^n] = \frac{1}{s}\left(\mathcal{L}[2t] + y(0)\right) = \frac{n!}{s^{n+1}}$$

(d) Theorem: Linearity of Laplace transform.

$$\mathcal{L}\left[f(t) + g(t)\right] = \mathcal{L}\left[f(t)\right] + \mathcal{L}\left[g(t)\right], \quad \mathcal{L}\left[cf(t)\right] = c\mathcal{L}\left[f(t)\right]$$

Proof: These are just properties of the definite integral.

$$\begin{aligned}\mathcal{L}[f+g] &= \int_0^\infty (f+g)e^{-st}\,dt \\ &= \int_0^\infty fe^{-st}\,dt + \int_0^\infty ge^{-st}\,dt \\ &= \mathcal{L}[f] + \mathcal{L}[g]\end{aligned}$$

Likewise for the second result.

(e) Expect to be able to prove these in an exam.

6. Example: We now see why Laplace transform will be useful for linear differential equations.

$$\frac{dy}{dt} + 5y = e^{-t}, \quad y(0) = 2$$

(a) Previous techniques:

- Direction field. See which curve is ours from the initial condition.
- Nonhomogenous equation. Solution is of the form

$$y = y_h + y_p$$

where y_h is the general solution to the homogeneous equation and y_p is a particular solution found via integrating factor.

$$\frac{dy}{dt} + 5y = 0 \quad \to \quad y_h = Ce^{-5t}$$

$$\frac{dy}{dt} + 5y = e^{-t} \quad \to \quad \frac{d}{dt}(e^{5t}y) = e^{5t}e^{-t} \quad \to \quad y_p = e^{-5t}\int e^{4t}\,dt = \frac{1}{4}e^{-t}$$

Using the initial condition, our solution is

$$y = y_h + y_p = Ce^{-5t} + \frac{1}{4}e^{-t} = \frac{7}{4}e^{-5t} + \frac{1}{4}e^{-t}$$

(b) Apply Laplace transform to both sides of the equation and denote $Y(s) = \mathcal{L}[y]$.

$$\begin{aligned}\mathcal{L}\left[\frac{dy}{dt} + 5y\right] &= \mathcal{L}\left[e^{-t}\right] \\ \mathcal{L}\left[\frac{dy}{dt}\right] + 5\mathcal{L}[y] &= \mathcal{L}\left[e^{-t}\right] \\ s\mathcal{L}[y] - y(0) + 5\mathcal{L}[y] &= \frac{1}{s+1}, \quad s > -1 \\ (s+5)\mathcal{L}[y] - 2 &= \frac{1}{s+1}, \quad s > -1 \\ (s+5)\mathcal{L}[y] &= 2 + \frac{1}{s+1}, \quad s > -1 \\ \mathcal{L}[y] &= \frac{2s+3}{(s+5)(s+1)}, \quad s > -1 \\ Y(s) &= \frac{2s+3}{(s+5)(s+1)}, \quad s > -1\end{aligned}$$

Notice we used no calculus here, just above two theorems.

(c) We found the transform $Y(s)$, now how to recover $y(t)$? Invert the Laplace transform.

7. Inverse Laplace transform:

   (a) You can show (though not easily) that the Laplace transform is invertible. That is for each transform $F(s)$, there is a unique $f(t)$ which it maps back to. There is a formula for this reverse mapping:
   $$f(t) = \mathcal{L}^{-1}[F] = \int_0^\infty e^{st} F(s)\, ds$$
   which just reverses the exponential multiplication. Lucky for us we have no need for this formula.

   (b) Note: Our favorite functions in this class are exponentials, sine/cosine, powers of t. Derivatives and integrals of these remain on the list. So if we know our basic Laplace transform formulas, we can just use them in reverse.

   (c) Theorem: Because $\mathcal{L}$ is a linear operator, $\mathcal{L}^{-1}$ is also linear. That is,
   $$\mathcal{L}^{-1}\left[F(s) + G(s)\right] = \mathcal{L}^{-1}\left[F(s)\right] + \mathcal{L}^{-1}\left[G(s)\right], \quad \mathcal{L}^{-1}\left[cF(s)\right] = c\mathcal{L}^{-1}\left[F(s)\right]$$

   (d) Example: Find the inverse Laplace transform of
   $$Y(s) = \frac{2s+3}{(s+5)(s+1)}.$$

   - Note, this function reminds us of
   $$\mathcal{L}[e^{at}] = \frac{1}{s-a}, \quad s > a$$
   - Use partial fraction decomposition to separate fraction into two parts.
   $$Y(s) = \frac{2s+3}{(s+5)(s+1)} = \frac{A}{s+1} + \frac{B}{s+5} = \frac{1/4}{s+1} + \frac{7/4}{s+5}$$
   Then,
   $$y(t) = \mathcal{L}^{-1}(Y(S)) = \mathcal{L}^{-1}\left(\frac{1/4}{s+1} + \frac{7/4}{s+5}\right) = \frac{1}{4}\mathcal{L}^{-1}\left(\frac{1}{s+1}\right) + \frac{7}{4}\mathcal{L}^{-1}\left(\frac{1}{s+5}\right) = \frac{1}{4}e^{-t} + \frac{7}{4}e^{-5t}$$
   This finishes the above example, same solution as before.
   - Note the use of linearity here.

8. Homework: 1-25, 27

.2 6.2: Discontinuous functions

1. Advantage of Laplace transform:

   (a) We have seen how Laplace transform can turn a differential equation into an algebra problem. All calculus is transferred into transform formulas and derivative / linearity theorems.

   (b) Here we will see a second advantage: ability to handle difficult functions
   - step functions / turn on a switch or new disease (this section)
   - delta (impulse) functions / shock or sudden impact (later)

2. Heaviside function:

   (a) Basic step function.
   $$H(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

(b) The switch can happen at any location $a > 0$ if we are considering $t \geq 0$.

$$H_a(t) = H(t-a) = \begin{cases} 1, & t \geq a \\ 0, & t < a \end{cases}$$

Graph and see. Looks like one side is heavier!

(c) Can even switch on and off as the unit box function:

$$H_{ab}(t) = H_a(t) - H_b(t)$$

(d) Multiplication by the Heaviside function allows you to switch on that function at location a. Graph:

$$y = f(t) = t^2 \quad \text{vs} \quad y = H_a(t)f(t)$$

(e) Notation: Text uses u_a (unit step function) in place of H_a (Heaviside function).

(f) Example: Forcing term only applied after $t = 2$.

$$\frac{dy}{dt} = -y + H_2(t)e^{-2(t-2)}, \quad y(0) = 1$$

Think of this as two differential equations:

$$\frac{dy}{dt} + y = \begin{cases} 0, & t < 2 \\ e^{-2(t-2)}, & t \geq 2 \end{cases}$$

where function e^{-2t} is switched on at $t = 2$.

3. Laplace transform of $H_a(t)$.

(a) Note, $H(t) = 1$ for $t \geq 0$ and we already have $\mathcal{L}[1]$ from last time. So,

$$\mathcal{L}[H(t)] = \frac{1}{s}, \quad s > 0$$

(b) Use $\mathcal{L}[H(t)]$ to compute $\mathcal{L}[H_a(t)]$.

$$\begin{aligned} \mathcal{L}[H_a(t)] &= \int_0^\infty H_a(t)e^{-st}\, dt \\ &= \int_a^\infty H(t-a)e^{-st}\, dt \\ &= \int_0^\infty H(\tilde{t})e^{-s(\tilde{t}+a)}\, d\tilde{t} \\ &= e^{-sa}\int_0^\infty H(\tilde{t})e^{-s\tilde{t}}\, d\tilde{t} \\ &= e^{-sa}\mathcal{L}[H(t)] \\ &= e^{-sa}\frac{1}{s} = \frac{e^{-sa}}{s} \end{aligned}$$

(c) Theorem (shift theorem): The above calculation generalizes. If $\mathcal{L}[f] = F(s)$, then

$$\mathcal{L}[H_a(t)f(t-a)] = e^{-as}F(s)$$

where $H_a(t)f(t-a)$ turns on f (for $t > 0$) starting at $t = a$. Draw picture to illustrate.

(d) Formula for $\mathcal{L}[f(t-a)]$ in terms of $\mathcal{L}[f(t)]$:

- Note above $\mathcal{L}[1] = \mathcal{L}[H] = \frac{1}{s}$. This poses a problem for $\mathcal{L}^{-1}$. Which one is it? Kind of the same since $t \geq 0$.
- Gets muddy when you shift general functions f to give $f(t-a)$. Information before zero was gone, now it is there. Our formula

$$\mathcal{L}[H_a(t)f(t-a)] = e^{-as}F(s)$$

says erase the information before zero.

4. Example: Forcing term only applied after $t = 2$.

$$\frac{dy}{dt} = -y + H_2(t)e^{-2(t-2)}, \quad y(0) = 1$$

(a) Apply Laplace transform. Denote $\mathcal{L}[y] = Y(s)$.

$$\begin{aligned}
\mathcal{L}[\frac{dy}{dt}] &= \mathcal{L}[-y + H_2(t)e^{-2(t-2)}] \\
s\mathcal{L}[y] - y(0) &= -\mathcal{L}[y] + \mathcal{L}[H_2(t)e^{-2(t-2)}] \\
sY(s) - 1 &= -Y(s) + e^{-2s}\mathcal{L}[e^{-2t}] \\
sY(s) - 1 &= -Y(s) + e^{-2s}\frac{1}{s+2} \\
(s+1)Y(s) &= 1 + e^{-2s}\frac{1}{s+2} \\
Y(s) &= \frac{1}{s+1} + \frac{e^{-2s}}{(s+1)(s+2)}
\end{aligned}$$

(b) Invert the Laplace transform.

$$\begin{aligned}
\mathcal{L}[y] &= \frac{1}{s+1} + \frac{e^{-2s}}{(s+1)(s+2)} \\
y &= \mathcal{L}^{-1}[\frac{1}{s+1}] + \mathcal{L}^{-1}[\frac{e^{-2s}}{(s+1)(s+2)}]
\end{aligned}$$

(c) Partial fraction decomposition:

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} + \frac{-1}{s+2}$$

(d) Finish inverting using shift theorem.

$$\begin{aligned}
y &= \mathcal{L}^{-1}[\frac{1}{s+1}] + \mathcal{L}^{-1}[\frac{e^{-2s}}{s+1}] - \mathcal{L}^{-1}[\frac{e^{-2s}}{s+2}] \\
&= e^{-t} + H_2(t)e^{-2(t-1)} - H_2(t)e^{-2(t-2)} \\
&= e^{-t} + H_2(t)(e^{-2(t-1)} - e^{-2(t-2)})
\end{aligned}$$

(e) Note, our piecewise differential equation gave a piecewise solution.

$$y(t) = \begin{cases} e^{-t}, & t < 2 \\ e^{-t} + e^{-2(t-1)} - e^{-2(t-2)}, & t \geq 2 \end{cases}$$

5. Homework: 1-17

.3 6.3: Second-order equations

1. Summary of Laplace transform results so far:
   (a) Definition: $\mathcal{L}[f(t)] = \int_0^\infty f(t)e^{-st}\,dt$
   (b) $\mathcal{L}[1] = \frac{1}{s}, s > 0$
   (c) $\mathcal{L}[e^{at}]\frac{1}{s-a}, s > a$
   (d) $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}, s > 0$
   (e) $\mathcal{L}[dy/dt] = s\mathcal{L}[y] - y(0)$, $\mathcal{L}[d^2y/dt^2] = s^2\mathcal{L}[y] - sy(0) - y'(0)$ (differentiation)
   (f) $\mathcal{L}[f+g] = \mathcal{L}[f] + \mathcal{L}[g], \mathcal{L}[cf] = c\mathcal{L}[f]$ (linearity)
   (g) $\mathcal{L}[H(t)] = \frac{1}{s}, s > 0$
   (h) $\mathcal{L}[H_a(t)] = \frac{e^{-sa}}{s}, s > 0$
   (i) $\mathcal{L}[H_a(t)f(t-a)] = e^{-as}\mathcal{L}[f(t)]$ (t-axis shift)
2. We have seen advantages of the Laplace transform (simpler process, handle new kinds of functions), but here we will see again the drawback of having to invert the Laplace transform.
3. Second order linear equation with forcing:
$$m\frac{dy}{dt^2} + b\frac{dy}{dt} + ky = f(t) \quad \Rightarrow \quad \frac{dy}{dt^2} + p\frac{dy}{dt} + qy = f(t)$$
where m is mass, b is damping, k is spring stiffness, f is applied external force.
   (a) Solutions include terms like $\sin(\omega t), \cos(\omega t), e^{at}\sin(\omega t), e^{at}\cos(\omega t)$. We need these Laplace tranforms.
   (b) Note, all of the four are of the form
$$e^{(a+ib)t} = e^{at}(\cos(bt) + i\sin(bt)$$
   and we know $\mathcal{L}[e^{at}]$ which holds even for complex numbers a.
   (c) Laplace transform of $\cos(bt)$. Use the fact that
$$\cos(bt) = \frac{e^{ibt} + e^{-bit}}{2}$$
   Then,
$$\mathcal{L}[\cos(bt)] = \frac{1}{2}\left(\mathcal{L}[e^{ibt}] + \mathcal{L}[e^{-bit}]\right) = \frac{1}{2}\left(\frac{1}{s-ib} + \frac{1}{s+ib}\right) = \frac{1}{2}\left(\frac{2s}{s^2+b^2}\right) = \frac{s}{s^2+b^2}$$
   (d) Laplace transform of $\sin(bt)$. Similar to $\cos(bt)$,
$$\mathcal{L}[\sin(bt)] = \frac{b}{s^2+b^2}$$
   (e) Multiplications of the form $e^{at}f(t)$ not surprisingly can be done straight from the definition of Laplace transform. Assume $\mathcal{L}[f] = F(s)$. Then,
$$\mathcal{L}[e^{at}f(t)] = \int_0^\infty e^{at}f(t)e^{-st}\,dt = \int_0^\infty f(t)e^{-(s-a)t}\,dt = F(s-a).$$
   Theorem: ($s-$axis shift) If $\mathcal{L}[f] = F(s)$. Then,
$$\mathcal{L}[e^{at}f(t)] = F(s-a)$$

(f) Example: Compute the inverse Laplace transform of

$$Y(s) = \frac{s+1}{s^2+6s+10}.$$

Need to complete the square of the denominator as $(s-\alpha)^2+b^2$ to see the shift and sine / cosine transform.

$$Y(s) = \frac{s+1}{s^2+6s+10} = \frac{s+1}{(s+3)^2+1} = \frac{s+3}{(s+3)^2+1} - 2\frac{1}{(s+3)^2+1}$$

Then,

$$\mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{s+3}{(s+3)^2+1}\right] - 2\mathcal{L}^{-1}\left[\frac{1}{(s+3)^2+1}\right] = e^{-3t}\cos(t) - 2e^{-3t}\sin(t)$$

Note we can now handle any proper rational function $\frac{p(s)}{q(s)}$ since can always factor polynomial q into a product of linear and irreducible quadratic factors (fundamental theorem of algebra).

4. Example:

$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 5y = 2e^t, \quad y(0) = 3, y'(0) = 1$$

(a) Laplace transform:

$$\mathcal{L}\left[\frac{d^2y}{dt^2}\right] - 4\mathcal{L}\left[\frac{dy}{dt}\right] + 5\mathcal{L}[y] = 2\mathcal{L}[e^t]$$
$$s^2Y(s) - sy(0) - y'(0) - 4(sY(s) - y(0)) + 5Y(s) = 2\frac{1}{s-1}$$
$$s^2Y(s) - 3s - 1 - 4sY(s) + 12 + 5Y(s) = \frac{2}{s-1}$$

(b) Solve for $Y(s)$:

$$(s^2-4s+5)Y(s) = 3s - 11 + \frac{2}{s-1}$$
$$Y(s) = \frac{3s-11}{(s-2)^2+1} + \frac{2}{(s-1)((s-2)^2+1)}$$

(c) Partial fraction decomposition:

$$Y(s) = \frac{1}{s-1} + \frac{2s-8}{(s-2)^2+1}$$

(d) Inverse Laplace transform:

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\left[\frac{1}{s-1}\right] + \mathcal{L}^{-1}\left[\frac{2s-8}{(s-2)^2+1}\right] \\ &= \mathcal{L}^{-1}\left[\frac{1}{s-1}\right] + 2\mathcal{L}^{-1}\left[\frac{s-2}{(s-2)^2+1}\right] - 4\mathcal{L}^{-1}\left[\frac{s-2}{(s-2)^2+1}\right] \\ &= e^t + 2e^{2t}\cos(t) - 4e^{2t}\sin(t) \end{aligned}$$

Note no need to solve for unknown constants for initial conditions via system of equations.

5. Homework: 1-31

.4 6.4: Delta functions and impulse forcing

1. We have seen how Laplace transform allows us to solve differential equations with new functions. Here we add to that list.
   (a) Discontinuous functions, turning on a switch (two sections ago)
   (b) Delta function, sudden impulse (this section)
2. Impulse forcing:
   (a) Example:
   $$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = f(t), \quad y(0) = 1, y'(0) = 1$$
   This is a spring mass equation with damping coefficient 2 and spring stiffness 5. f is an external force. Here we want to understand what happens if f is a sudden impact, a kick to the mass on the spring at a certain time t.
   (b) Unit impulse at zero: Approach this idea over smaller and smaller intervals.
   - $f(t) = F$ on the interval from $[a, b]$. A unit force is such that
   $$\int_{-\infty}^{\infty} f(t)\ dt = F(b-a) = 1$$
   - At zero: $f(t) = \frac{1}{h}$ over the interval $[0, h]$. Then
   $$f(t) = \frac{1}{h}\left[H(t) - H(t-h)\right] \quad \text{where } H \text{ is the Heaviside function}$$
   and
   $$\int_{-\infty}^{\infty} f(t)\ dt = \frac{1}{h}h = 1$$
   The smaller h is, the taller and skinnier this step function is.
   - Taking the limit as $h \to 0$
   $$\lim_{h\to 0} \frac{1}{h}\left[H(t) - H(t-h)\right]$$
   gives the *Dirac delta function*.
   $$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$
   such that
   $$\int_{-\infty}^{\infty} \delta(t)\ dt = 1$$
   - $\delta(t)$ is not really a function like we are used to (taking ∞ as an output makes no sense), but it is useful when the Laplace transform comes in.
   - Also note the appearance of the difference quotient leading to the fact that
   $$H'(t) = \delta(t).$$
   (c) Laplace transform of the Delta function:
   $$\mathcal{L}\left[\frac{1}{h}\left[H(t) - H(t-h)\right]\right] = \frac{1}{h}\left[\frac{1}{s} - \frac{e^{-hs}}{s}\right] = \frac{1}{s} - \frac{e^{-hs}}{hs} \to 1$$
   via lHospital's rule. Then,
   $$\mathcal{L}\left[\delta(t)\right] = 1$$
   and we see a hint why the Delta function is important.

(d) Modified forcing:

- Unit forcing at time a: $\delta_a(t) = \delta(t-a)$ (just a horizontal shift so we know the Laplace transform)
$$\mathcal{L}[\delta_a(t)] = e^{-as}$$
- Forcing of strength A: $A\delta(t)$ (again a bit strange with the infinite value at $t=0$, but the Laplace transform shows the strength change)
$$\mathcal{L}[A\delta(t)] = A$$

3. Example: This basic example illustrates the idea of impulse force.
$$\frac{d^2y}{dt^2} + y = A\delta_{\pi/2}(t), \quad y(0) = 1, y'(0) = 0$$
Spring-mass with no damping. Impulse force of strength A is applied at time $\pi/2$, the equilibrium point.

(a) Laplace transform:
$$s^2Y - s + Y = Ae^{-\frac{\pi}{2}s} \quad \Rightarrow \quad Y = \frac{s}{s^2+1} + \frac{Ae^{-\frac{\pi}{2}s}}{s^2+1}$$

(b) Inverse Laplace transform:
$$y(t) = \cos(t) + H(t-\pi/2)A\sin(t-\pi/2) = \begin{cases} \cos(t), & 0 \le t \le \frac{\pi}{2} \\ (1-A)\cos(t), & t \ge \frac{\pi}{2} \end{cases}$$

(c) Check that the final solution (displacement) is continuous as it should be.

(d) Graph in Desmos with slider for A. `https://www.desmos.com/calculator/7saayio0fw`

4.
$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = \delta_3(t), \quad y(0) = 1, y'(0) = 1$$

(a) Laplace transform:
$$s^2Y - sy(0) - y'(0) + 2sY - y(0) + 5Y = e^{-3s} \quad \rightarrow \quad Y = \frac{2 + s + e^{-3s}}{s^2 + 2s + 5}$$

(b) Inverse Laplace transform:
$$y = \mathcal{L}^{-1}\left[\frac{2+s+e^{-3s}}{(s+1)^2+4}\right] = \mathcal{L}^{-1}\left[\frac{s+1}{(s+1)^2+4} + \frac{1}{2}\frac{2}{(s+1)^2+4} + \frac{1}{2}\frac{2e^{-3s}}{(s+1)^2+4}\right]$$
and then
$$y = e^{-t}\cos(2t) + \frac{1}{2}e^{-t}\sin(2t) + \frac{1}{2}H_3(t)e^{-(t-3)}\sin(2(t-3))$$
which is more cleanly written as
$$y = \begin{cases} e^{-t}\cos(2t) + \frac{1}{2}e^{-t}\sin(2t), & 0 \le t \le 3 \\ e^{-t}\cos(2t) + \frac{1}{2}e^{-t}\sin(2t) + \frac{1}{2}e^{-(t-3)}\sin(2(t-3)) & t \ge 3 \end{cases}$$

5. Homework: 1-7

.5 6.5: Convolutions

1. Here we develop a new idea (convolution of two functions $f * g$) which is useful in two ways:
   (a) To help analyze inverse Laplace transforms of products.
   $$\mathcal{L}^{-1}[F(s)G(s)]$$
   (b) To derive a formula for any second order linear equation with forcing.
   $$\frac{d^2y}{dt^2} + p\frac{dy}{dt} + qy = f(t)$$
   (c) Bonus: Idea of convolution applies to a much more general setting than simply a tool for Laplace transform.
2. Idea: Laplace transform of products are difficult.
   (a) $\mathcal{L}[f(t)g(t)]$ has no nice formula in general.
   (b) We will derive a nice formula for $\mathcal{L}^{-1}[F(s)G(s)]$ as
   $$FG = \int_0^\infty (f * g)e^{-st}\ dt \quad \Rightarrow \quad \mathcal{L}^{-1}[FG] = f * g.$$
   (c) Motivation: Recall we motivated Laplace transform as a continuous analog of a power series.
   $$p(x) = \sum_{n=0}^{\infty} a_n x^n \quad \Rightarrow \quad F(s) = \int_0^{infty} f(t)e^{-st}\ dt$$
   where we rewrote x^t as e^{-st}. For two power series
   $$p(x) = \sum a_n x^n, \quad q(x) = \sum b_n x^n$$
   there is no nice way to find
   $$\sum (a_n b_n) x^n$$
   in terms of general p, q. Though, there is a way to find
   $$p(x)q(x) = \left(\sum a_n x^n\right)\left(\sum b_n x^n\right) = \sum c_n x_n$$
   called the Cauchy product of a power series. Its name is a *discrete convolution* and the formula is
   $$c_n = \sum_{i=0}^{n} a_{n-i} b_i.$$
   Offer bonus bounty to derive this discrete formula for convolution.
3. Convolution formula: This is a strange formula at first glance, but the same can be said for the Laplace transform.
   $$f(t) * g(t) = \int_0^t f(t-u)g(u)\ du$$
   is the new function of t such that
   $$\mathcal{L}[f * g] = \mathcal{L}[f]\mathcal{L}[g] = F(s)G(s)$$
   (will show this soon) and is useful when calculating the inverse transform of products
   $$\mathcal{L}^{-1}[FG].$$

(a) Result: The convolution acts like a product in the sense that

$$f(t) * g(t) = g(t) * f(t).$$

This result is not clear at first glance thru the formula (need a substitution of $v = t - u$), but if you trust the Laplace transform result added to the uniqueness of the transform,

$$\mathcal{L}[f * g] = F(s)G(s) = G(s)F(s) = \mathcal{L}[g * f]$$

we see that this is just regular real number multiplication which we know is communative.

(b) Example: Let $f(t) = t^2$ and $g(t) = t$.

$$t * t^2 = \int_0^t (t-u)u^2 \, du = \int_0^t tu^2 - u^3 \, du = t\frac{u^3}{3} - \frac{u^4}{4}\Big|_0^t = \frac{t^4}{12}$$

Note

$$\mathcal{L}[t * t^2] = \mathcal{L}[t^4/12] = \frac{1}{12}\frac{4!}{s^5} = \frac{2}{s^5}.$$

But,

$$\mathcal{L}[t]\mathcal{L}[t^2] = \frac{1}{s^2}\frac{2}{s^3} = \frac{2}{s^5}$$

and our formula holds at least in this case.

(c) Example: This product is strange though.

$$1 * f(t) = \int_0^t f(u) \, du$$

Convolution with 1 is the antiderivative of f.

(d) Example: The convolution identity is $\delta(t)$.

$$\delta * f(t) = 1$$

4. Convolution formula derivation:

(a) Denote $F = \mathcal{L}[f]$ and $G = \mathcal{L}[g]$. Then we want a formula for $\mathcal{L}^{-1}[FG]$.

$$\begin{aligned} F(s)G(s) &= \left[\int_0^\infty f(\tau)e^{-s\tau} \, d\tau\right]\left[\int_0^\infty g(u)e^{-su} \, du\right] \\ &= \int_0^\infty \int_0^\infty f(\tau)g(u)e^{-s(\tau+u)} \, du \, d\tau \end{aligned}$$

This is a double integral over the first quadrant of the $u\tau$-plane. Again, our goal is a single Laplace transform of the mold

$$F(s)G(S) = \int_0^\infty h(t)e^{-st} \, dt.$$

Do a change of variable calling $t = \tau + u$, $u = u$. Then,

$$\begin{aligned} F(s)G(s) &= \int_0^\infty \int_0^\infty f(\tau)g(u)e^{-s(\tau+u)} \, du \, d\tau \\ &= \int_0^\infty \int_0^t f(t-u)g(u)e^{-st} \left|\frac{\partial(u,\tau)}{\partial(u,t)}\right| \, du \, dt \\ &= \int_0^\infty \int_0^t f(t-u)g(u) \, du \, e^{-st} \, dt \\ &= \int_0^\infty f(t) * g(t) \, e^{-st} \, dt \end{aligned}$$

Note the Jacobian is computed as the following determinant.

$$\left|\frac{\partial(u,\tau)}{\partial(u,t)}\right| = \begin{vmatrix} \partial u/\partial u & \partial u/\partial t \\ \partial \tau/\partial u & \partial \tau/\partial t \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1$$

begin that $u = u$ and $\tau = t - u$. Also the first quadrant of the $u\tau$ plane $(0 < u < \infty, 0 < \tau < \infty)$ is achieved by u, t by dragging lines of slope negative 1 ($t = \tau + u$, $0 < u < t$) across the first quadrant in a diagonal fashion ($0 < t < \infty$).

5. Example: It turns out this convolution formula is not all too practical. Compute the inverse Laplace transform of

$$F(s)\frac{3s}{(s^2+1)(s^2+4)}.$$

   (a) Convolution solution:

$$F(s)\frac{3s}{(s^2+1)(s^2+4)} = 3\frac{s}{s^2+1}\frac{s^2+4}{=}3G(s)H(s)$$

   Noting that $g(t) = L^{-1}[G] = \sin(t)$ $h(t) = L^{-1}[H] = \cos(2t)$, then

$$\mathcal{L}^{-1}[F] = 3g(t) * h(t) = 3\int_0^t g(t-u)h(u)\ du = 3\int_0^t \sin(t-u)\cos(2u)\ du$$

   which requires the a bit of trig integration.

   (b) Solution by partial fractions:

$$f(t) = \cos(t) - \cos(2t)$$

   so this must match the above definite integral (convolution $g * f$).

6. Convolution solution to general second order equation:

$$\frac{d^2y}{dt^2} + p\frac{dy}{dt} + qy = f(t), \quad y(0) = y'(0) = 0$$

   where f is a general forcing function.

   (a) Take Laplace transform:

$$s^2Y + psY + qY = F \quad \Rightarrow \quad Y = F\frac{1}{s^2+ps+q} = FW$$

   (b) $\mathcal{L}[y] = Y$ is a product $F(s)W(s)$ making y a convolution of the forcing function f with w.

$$y = f * w = \int_0^t f(t-u)w(u)\ du$$

   How to interpret this?

   (c) What is w?

   - $W(s) = \frac{1}{s^2+ps+q}$ is called the *transfer function* for the system.
   - $w(t)$ is the *weight function* for the system.
   - How to interpret w? Note that $\mathcal{L}[\delta] = 1$. Then,

$$y'' + py' + qy = \delta, \quad y(0) = y'(0) = 0 \quad \Rightarrow \quad Y = \frac{1}{s^2+ps+q} = W(s)$$

     giving $y = w$. We kick the mass starting at rest from equilibrium with unit impulse. The solution here is the response to this unit kick.

- Then,

$$y = f * w = \int_0^t f(t-u)w(u)\ du$$

is the combination of all these unit kicks as governed by f. Kick, kick, kick until accumulate solution y.

7. Bonus convolution idea:
   (a) $f * g$ tells us to what degree f and g overlap across the domain of t.
   (b) How the shape of one is modified by the other.
   (c) `https://en.wikipedia.org/wiki/Convolution`
8. Homework: 1-9

.6 6.6: The qualitative theory of Laplace transform